

OPENCUA: Open Foundations for Computer-Use Agents

Xinyuan Wang ^{*x} Bowen Wang ^{*x} Dunjie Lu ^{*x} Junlin Yang ^{*x} Tianbao Xie ^{*x} Junli Wang ^{*x}
Jiaqi Deng ^x Xiaole Guo ^x Yiheng Xu ^x Chen Henry Wu ^c Zhennan Shen ^x Zhuokai Li ^x Ryan Li ^x
Xiaochuan Li ^x Junda Chen ^x Boyuan Zheng ^x Peihang Li ^x Fangyu Lei ^x Ruisheng Cao ^x Yeqiao Fu ^x
Dongchan Shin ^x Martin Shin ^x Jiarui Hu ^x Yuyan Wang ^x Jixuan Chen ^x Yuxiao Ye ^x Danyang Zhang ^x
Dikang Du ^m Hao Hu ^m Huarong Chen ^m Zaida Zhou ^m Yipu Wang ^m Heng Wang ^m
Diyi Yang ^s Victor Zhong ^w Flood Sung ^m Y.Charles ^m Zhilin Yang ^m Tao Yu ^{†x}

^x XLANG Lab, University of Hong Kong ^m Moonshot AI

^s Stanford University ^w University of Waterloo ^c Carnegie Mellon University

OpenCUA Homepage (Tool, Model, Data): <https://opencua.xlang.ai>

Abstract

Vision-language models have demonstrated impressive capabilities as computer-use agents (CUAs) capable of automating diverse computer tasks. As their commercial potential grows, critical details of the most capable CUA systems remain closed. As these agents will increasingly mediate digital interactions and execute consequential decisions on our behalf, the research community needs access to open CUA frameworks to study their capabilities, limitations, and risks. To bridge this gap, we propose OPENCUA, a comprehensive open-source framework for scaling CUA data and foundation models. Our framework consists of: (1) an annotation infrastructure that seamlessly captures human computer-use demonstrations; (2) AGENTNET, the first large-scale computer-use task dataset spanning 3 operating systems and 200+ applications and websites; (3) a scalable pipeline that transforms demonstrations into state-action pairs with reflective long Chain-of-Thought reasoning that sustain robust performance gains as data scales. Our end-to-end agent models demonstrate strong performance across CUA benchmarks. In particular, OPENCUA-32B achieves an average success rate of **34.8%** on **OSWorld-Verified**, establishing a new state-of-the-art (SOTA) among open-source models and surpassing OpenAI CUA (GPT-4o). Further analysis confirms that our approach generalizes well across domains and benefits significantly from increased test-time computation. We release our annotation tool, datasets, code, and models to build open foundations for further CUA research: <https://opencua.xlang.ai>

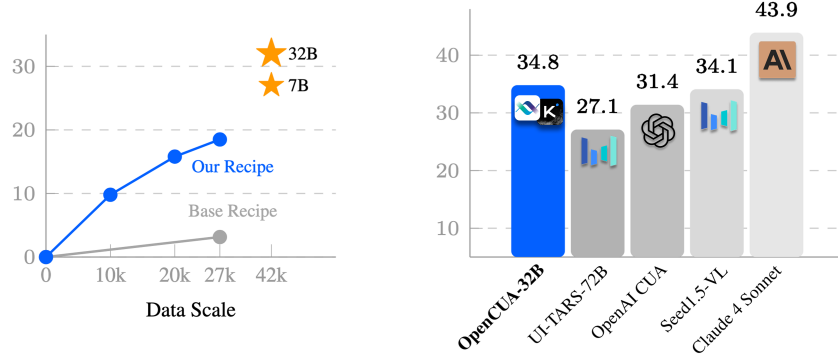


Figure 1: **Left:** OpenCUA recipe helps performance scaling with data size and model size. **Right:** Performance on **OSWorld-Verified** [42]: OpenCUA-32B outperforms current open-source models and is comparable with OpenAI CUA (GPT-4o based) [28].

*Equal contribution. †Corresponding authors.

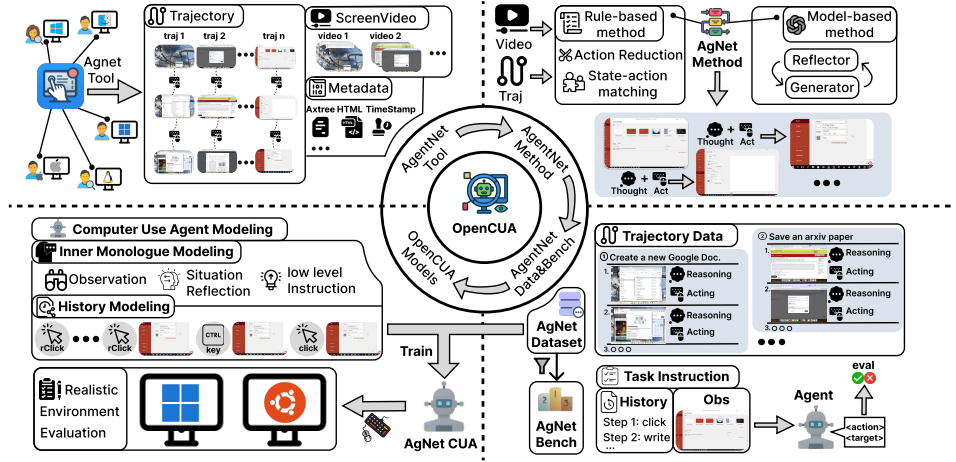


Figure 2: Overview of the OPENCUA framework. **Top left:** AGENTNET TOOL captures user interactions across OSes with screen videos and action flows. **Top right:** Raw demos are processed into state-action trajectories with reasoning and history. **Bottom right:** AGENTNET Dataset and Bench provide diverse tasks and offline evaluation with gold-standard actions. **Bottom left:** OPENCUA models are trained and able to execute in realistic environments.

1 Introduction

Computer-use agents (CUAs), powered by vision-language models (VLMs), aim to autonomously complete computer tasks and have great potential in facilitating daily and professional workflows. Despite their growing role in high-stakes decision-making, critical details including training data, architectures, and development processes about how state-of-the-art CUA systems are built remain closed and proprietary [3, 4, 5, 17, 28]. As the lack of transparency limits technical advancements and raises safety concerns [31, 38, 11], the research community needs *truly open* CUA frameworks to study their capabilities, limitations, and risks.

However, current open-source attempts in CUA face significant challenges that impede progress. Firstly, there is no open-source scalable infrastructure for collecting diverse large-scale computer-use data - a complex requirement that involves the real-time capture of user interactions and state information, followed by transformation into agent-executable trajectories. Secondly, existing open-source graphical user interface (GUI) datasets remain limited in scope and scale due to the complexity and high cost of data collection; they either focus on specific domains (grounding [10, 41, 18], mobile [30, 24], or web [14, 12]) or lack sufficient diversity for general computer-use applications. Furthermore, many CUA works provide insufficient details about their modeling strategies and training recipes, making replication difficult even with access to their collected data. These limitations collectively hinder advances in general-purpose CUAs and restrict a meaningful exploration of their scalability, generalizability, and potential learning approaches.

To address these challenges, we introduce OPENCUA, a fully open-source framework to scale the CUA data and the foundation models (Figure 2). To address infrastructure challenges, we first develop a user-friendly, cross-OS computer task annotation application AGENTNET TOOL that can be installed on personal computers to seamlessly record natural human demonstrations and corresponding computer states, without disrupting the user’s workflow (Figure 2 top left). We then collect the AGENTNET dataset, including more than 20K open-domain computer task trajectories spanning over 100 applications and 200 websites across Windows, macOS, and Ubuntu (Figure 2 top right). This dataset authentically captures the complexity of human behaviors and environmental dynamics from users’ personal computing environments. Furthermore, given that online CUA benchmarks such as OSWorld [42] require substantial environment setup effort and runtime resources, we curated AGENTNETBENCH based on our collected human demonstrations (Figure 2 bottom right). This offline benchmark provides multiple gold-standard actions per step, efficiently approximating online metrics to dramatically accelerate agent evaluation and development.

Critical to our OPENCUA framework is our (1) data processing pipeline and (2) novel modeling and training recipe for constructing CUA training data from human demonstrations. We first introduce an

action discretization pipeline that converts raw human demonstrations, which typically consist of videos and high-frequency, redundant keyboard/mouse actions, into state-action pairs feasible for vision language model training. Despite this, we observe that training on state-action pairs alone yields limited performance gains even as the dataset size scales (see Figure 1 Left). Our first key insight is that scaling agent capabilities requires augmenting these trajectories with reflective long Chain-of-Thought (CoT) reasoning. We propose a reflective CoT synthesis method that explicitly injects planning, memory, and reflection into the per-step reasoning process through natural language “inner monologue” (Section 3.1). Different from previous work, our reasoning traces are notably more detailed and contain reflection thoughts that help the agent detect and recover from errors. Moreover, we identify key modeling details that improve agent performance (Section 3.2), such as multi-image history. Finally, we show that carefully designing training data mixtures—including diverse reasoning and general text—is beneficial for computer-use agent training (Section 3.3).

Built upon our methodology, we developed strong computer-use agent models through supervised fine-tuning (SFT) (Figure 2 bottom left). Our results show that our approach enables robust performance scaling with increased data size (Section 4.2). Our model, OPENCUA-32B, achieves a success rate of **34.8%** (100 step) on **OSWorld-Verified** [42, 45], establishing a new state-of-the-art among the open-source models, even outperforming the proprietary OpenAI CUA (GPT-4o based) [28]. We did extensive experiments and analysis on various model structures and data scales in Section 5. Because of the diversity and coverage of our training data, our models show strong cross-domain generalization. Our agent models also show promising scalability with increased test-time compute, such as increased number of steps and larger n in Pass@ n evaluation. We also did additional experiments, including grounding and robustness analysis. Finally, we also provide detailed ablations to justify the important design choices in our method and training recipe (Section 5). We open-source the complete suite of our OPENCUA framework, including the annotation tool, collected datasets, code, benchmark and models, providing open foundations for further CUA research.

2 AGENTNET Collection

OPENCUA aims to scale desktop computer-use data across diverse computer environments and user scenarios. We prioritize collecting demonstrations that follow natural user behavior, imposing the least additional constraints on how users interact with their computers to improve the scalability of data collection. To this end, we developed **AGENTNET TOOL** and collected **AGENTNET** dataset, the first large-scale desktop agent task dataset.

2.1 Task Definition

Human Action	Action Description	Agent Action
Click	Click at a specific position	click ($x, y, button$)
Middle Click	Middle click at a specific position	middleClick (x, y)
Double Click	Double click at a specific position	doubleClick ($x, y, button$)
Triple Click	Triple click at a specific position	tripleClick ($x, y, button$)
Mouse Move	Move mouse to a specific position	moveTo (x, y)
Drag	Drag mouse from one position to another	dragTo (x, y)
Scroll	Scroll vertically or horizontally	scroll (dx, dy) / hscroll (dx, dy)
Type	Type a string of text	write (text)
Press	Press a specific key	press (key)
Hotkey	Perform a combination of keys	hotkey (key1, key2)
Terminate	End the task with success or failure	terminate (‘success’ or ‘fail’)

Table 1: Overview of Human Actions and Corresponding Agent Action Functions

We model the agent’s decision-making process – iterative observation of the computer state followed by action prediction – as a state-action transition trajectory: $(I, \langle s_0, a_0 \rangle, \langle s_1, a_1 \rangle, \dots, \langle s_T, a_T \rangle)$. Given a task language instruction I and initial state s_0 , the agent sequentially predicts a action a_i until goal state s_t and performs the termination action a_T : $P(a_i|I, s_0, a_0, \dots, s_i)$.

An important design choice in building computer-use agent is to convert compute state s_i into model observation. In this work, we follow the recent trend of building pure vision-based computer agents [29, 48, 41] and use the screenshot of the computer as the observation for the agent. We use human

computer-use actions, including keyboard and mouse movements, as the action space. To ensure the action space is applicable across various operating systems, we select a subset of PyAutoGUI actions and augment them with several necessary agent actions including the ‘success’ and ‘fail’ termination actions. The complete action space and its parameters are listed in Table 1.

2.2 AGENTNET TOOL

Efficient and accurate annotation is essential for collecting high-quality computer-use agent data, yet no existing tools support natural, cross-platform task recording by non-technical users. To address this, we developed a user-friendly annotation tool that streamlines the collection and verification of computer-use demonstrations (Figure 3), runs on annotators’ personal computers and records demonstrations in the background, capturing: (1) screen videos, (2) mouse and keyboard signals, and (3) accessibility trees (Axtree). These data are then processed

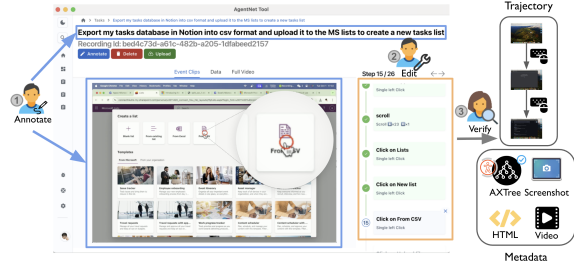


Figure 3: AGENTNET TOOL annotation and verification.

into state-action trajectories (see details below in Section 2.2), allowing annotators to review, edit, and submit demonstrations along with task instructions describing the overall goal. Former works require the annotators to demonstrate “gold” trajectories with all-correct steps, but this actually limits model’s capability to detect and recover from errors. We believe that annotation error is not all bad, as long as we can identify and utilize them (see Section 3.1), so we relax the requirement of all correct actions. Our implementation leverages several established tools: mouse and keyboard input tracking is based on DuckTrack [34] and OpenAdapt [27]; screen recording utilizes OBS Studio [26]; and accessibility tree (Axtree) parsing follows the OSWorld framework [42]. Additional implementation details can be found in Appendix C.3.1.

Annotation pipeline We designed our data collection with two key goals: **diversity** and **complexity**. Annotators were provided a curated list of around 200 applications and websites spanning various domains and were encouraged to demonstrate complex workflows involving professional features or multi-app interactions. Tasks were required to have more than 15 steps; those with <5 steps were rejected. To ensure wide coverage and real-world authenticity, we recruited annotators from both crowd-sourcing platforms and annotation companies. All annotators signed consent forms, and we use a multi-layer privacy protection mechanism to safeguard user data (Appendix C.3.2). To study model generalization, we split data into Windows/macOS and Ubuntu, ensuring no overlap with OSWorld tasks to prevent data leakage. All tasks were manually verified and labeled as *rejected*, *ok*, *good*, or *excellent* based on goal clarity, diversity, and complexity. Other annotation details are provided in Appendix C.2.

Constructing compact state-action trajectories Raw demonstrations consist of high-frequency screen recordings and fine-grained interaction signals (mouse movements, clicks, scrolls, key presses). A typical task can produce thousands of low-level actions that are too dense and inefficient for training. To address this challenge, we developed techniques including action reduction and state-action matching to construct compact state-action pairs $\langle s_i, a_i \rangle$. **(1) Action reduction:** We developed a rule-based method to compress and reduce these dense action signals into a smaller set of meaningful actions while preserving essential action information. We first compress atomic signals into higher-level operations. Mouse move events are treated as preconditions for clicks or drags, and only their start and end positions are retained. Scrolls are merged into single-directional actions with accumulated wheel counts. Consecutive key presses are merged into text input strings, while modifier combinations (e.g., CTRL+C) are abstracted into hotkey actions. We also combine common multistep gestures such as drags or double-clicks. This process yields a streamlined action sequence aligned with the pyautogui action space, as shown in Table 1. **(2) State-action matching:** To pair each action a_i with a representative state s_i , we extract keyframes from the screen recording that capture the system state *immediately before* the action occurs. However, naively aligning keyframes to action timestamps of mouse clicks risks leaking future information; e.g., the mouse may already be positioned over a button, making the prediction trivial. To address this challenge, for mouse

Table 2: Comparison between OPENCUA and Other GUI Datasets

Dataset	Tasks	Avg. Step	Env. Type	Personalized Env.	Human Traj.	Dom/ AxTree	Video	Inner Monologue
AndroidControl[22]	15283	5.5	Mobile	✗	✓	✓	✗	Short
AMEX[8]	2991	11.9	Mobile	✗	✓	✗	✗	✗
AitW[30]	2346	8.1	Mobile	✗	✓	✓	✗	✗
AitZ[52]	1987	6.0	Mobile	✗	✓	✗	✗	Short
GUI Odyssey[24]	7735	15.3	Mobile	✗	✓	✗	✗	✗
WonderBread[37]	598	8.4	Web	✗	✓	✓	✓	✗
AgentTrek[46]	10398	12.1	Web	✗	✗	✓	✓	Short
Mind2Web[12]	2350	7.3	Web	✗	✓	✓	✗	✗
GUIAct[9]	2482	6.7	Web	✗	✓	✓	✗	✗
AgentNet	20631²	17.2	Desktop	✓	✓	✓	✓	Long

clicks, we backtrack to the beginning of the mouse’s pre-movement phase and search backward to find the last visually distinct frame. After the final action, we append a terminal frame along with a corresponding termination action.

2.3 AGENTNET Statistics

Our dataset consists of 20,631 human-annotated computer-use tasks, including 12K from Windows, 5K from macOS and 3K from Ubuntu, with screen resolutions ranging from 720p to 4K. Each trajectory averages 17.2 steps, reflecting task complexity. As shown in Figure 4, the data spans over 140 applications and 190 websites, often involving multi-app workflows, professional tools, and uncommon features. Compared to prior GUI datasets (Table 2), AGENTNET is the first desktop trajectory-level dataset that is realistic, complex, diverse, and multi-modal. Full statistics are provided in Appendix C.1.

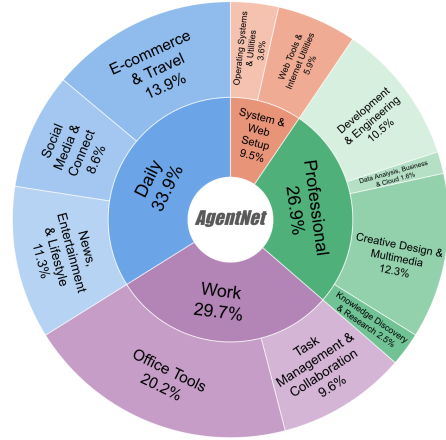


Figure 4: Domain distribution of tasks in AgentNet dataset

3 Training Computer-Use Agent Model

Our AGENTNET consist of task instructions I and state-action $\langle s_i, a_i \rangle$ trajectories. However, we find that directly finetuning vision-language action (VLA) models on our 27K trajectories leads to poor performance (4.4% success rate on OSWorld [42], shown in Figure 1 Left "Base Recipe"). This section presents our modeling and training recipe to enable scalable training of computer-use agent models, including a novel reasoning augmentation, context encoding, and data mixtures techniques.

3.1 Synthesizing Reflective Long CoT Reasoning

Consistent with prior works [51, 48, 29], we find natural language reasoning crucial for generalizable computer-use foundation models, helping CUAs internalize cognitive capabilities. We propose a multi-stage CoT framework synthesizing structured reasoning per state-action pair $\langle s_i, a_i \rangle$. Inspired by Aguis [48], our structured CoT includes three reasoning levels. The hierarchy begins with **L3**, contextual observation capturing salient visual and textual elements. Next, **L2** provides reflective reasoning analyzing state transitions, recalling previous steps, correcting errors, and planning subsequent actions. Finally, the model predicts **L1**, a concise executable action grounded in prior perception and thought. This $L3 \rightarrow L2 \rightarrow L1$ structure mirrors perceptual-to-agentic decision flow, equipping the model with coherent, interpretable reasoning trajectories that enhance generalization and robustness.

Reflection augmentation for L2 reasoning Incorrect or redundant annotations in human demonstrations are not all bad, as long as we can identify and use them to teach the identification and

²A total of 41,428 trajectories were used to train OPENCUA-7B and OPENCUA-32B, and 27,804 trajectories were used for training OPENCUA-QWEN2-7B and OPENCUA-A3B. Moonshot AI annotated the Ubuntu subset and generously agreed to release 3K of the annotated trajectories to the public.

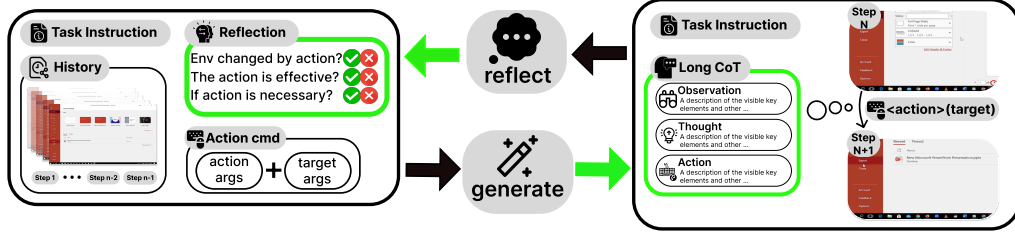


Figure 5: Reflective long CoT synthesis pipeline: generator and reflector iteratively generate and verify the reasoning components between the observation and ground-truth actions.

Task: In the "Features" slide, insert a table with 5 rows and 2 columns.

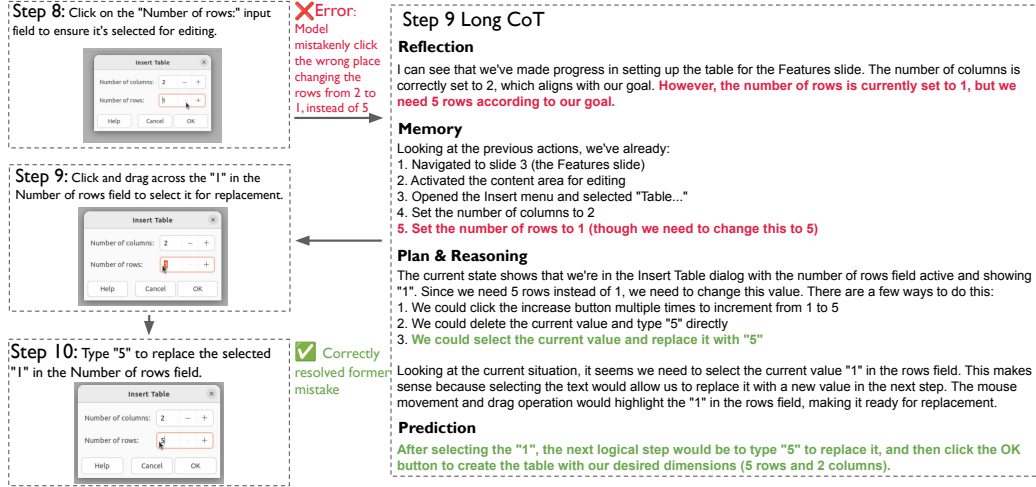


Figure 6: Reflective Long CoT Example: Before predicting the actual action, the model first reason according to the history and current action with reflection, memory, plan&reasoning and prediction in the CoT. The model identifies the former mistake and correct it in the later steps.

correction of model errors. Therefore, we designed a reflector to identify errors and generate reflection reasoning for each step. Our CoT synthesis framework extends the pipeline of Aguvis [48] and ActRe [50] by equipping the "Thought" with more comprehensive agent components, especially state-transit perception and reflection, and minimizing hallucination. As shown in Figure 5, our CoT synthesis pipeline consists of three components: reflector, generator, and summarizer. The reflector inspects each step for correctness and redundancy by comparing screenshots before and after the action, examining the correctness of the action code itself and the generated CoT, especially whether the "Action" aligns with the screenshot and code. When the step is incorrect or redundant, the reflector will elaborate reason and this step will be ignored during training. If the step is correct, the reflector will explain the differences the actions brings to the before and after state. The generator conditions on the full agent context—previous reflections, action history, task goal, screenshots, and action code—to generate structured CoT. To help the model ground coordinate-related actions more accurately, we incorporate *visual cues*: a red marker on the mouse action coordinate and a zoomed-in image patch (inspired by V* [39]). Finally, the summarizer refines vague user-written goals into more precise and aligned task objectives, and scores each trajectory for alignment, efficiency, and difficulty. Our method produces rich and meaningful CoTs that significantly improve model reasoning and planning. We use `claude-3-7-sonnet-20250219` as the base model for synthesizing the three components. The reflection helps agent model identify former errors and adjust future plan to make the task back to the right track. An example of error identification and correction in evaluation can be seen in Section F. Ablations in Section 5 demonstrate that this module is a important driver of performance gains.

3.2 Context Encoding and Test-Time Reasoning

For end-to-end agent models, history encoding plays a critical role in reasoning and planning.

Context encoding: (1) Textual history: We propose a structured inner monologue framework for textual history representation. Specifically, we adopt a dialogue-style history format representing the model’s responses and corresponding computer screenshots. Following Aguvis [48], we use L1 CoT (Action) to represent previous steps because it is more token-efficient and allows longer history windows without truncation. Moreover, our inner monologue includes memory components, further compensating for the absence of richer CoT in earlier steps. See Section 5 for ablations on history representation. (2) Visual history: Multi-image screenshot history is essential for agent model performance because screenshots are lossless visual representation of history, providing more grounded context than textual summaries. However, including more images also increases input length and reduces training efficiency. By default, we use three screenshots as visual representation, as our experiments show that this achieves a balance between performance and efficiency (see Figure 11).

Test-time reasoning format: While the model is trained with a mixture of CoT levels, we adopt the L2 CoT format at inference time due to its richer reasoning content, which enhances the model’s ability to reflect, plan, and reason. As shown in our ablation studies (Section 5), L2 CoT significantly improves test-time performance scalability—Pass@ n success rates on OSWorld increase markedly over Pass@1. In contrast, models lacking this reasoning augmentation exhibit limited scalability, highlighting the importance of strong reasoning signals at inference time.

3.3 Training Data Mixtures

CoT data mixture: As we mentioned in the Section 3.1, our structured inner monologue contains three levels of CoT: L1 (Action), L2 (Thought + Action), and L3 (Observation + Thought + Action), each encoding complementary information for agent decision-making but has different conceptual information. L1 CoT has direct connection to the actual action; while there is helpful screenshot perception information in the L3 CoT, some irrelevant elements may also be described; L2 CoT contains planning and prediction that directly affect the predicted action in L1. Therefore, we propose to train the model with a mixture of all three levels of CoT to reinforce this different levels of connection. Data example of L1, L2, and L3 can be seen in Appendix G. We verify this design choice with ablations in Section 5.

Mixture of grounding, planning, and general SFT data: A general-purpose computer-use agent foundation model should be capable of both solving complex computer-use tasks and performing general reasoning grounded in world knowledge. To achieve this, we train on a mixture of data types that span both computer-use and general vision-language domains.

For grounding, we initialize the model using existing datasets such as ShowUI [23], UGround [15], and 189K bounding-box samples parsed from collected AXTree structures. For planning and reasoning, we include a diverse mix of Ubuntu and Windows/macOS demonstrations as well as task-instruction-augmented samples (Section 3.1). To enhance generalization and reasoning ability, we additionally incorporate general supervised fine-tuning (SFT) data from the Kimi Team [33]. The general text data covering domains such as instruction following, mathematical reasoning, and long-context understanding. The general vision data includes domains such as OCR and vision QA data. This mixture ensures both GUI grounding and high-level reasoning capabilities across domains. Although these general data are not directly related to GUI environments, we find that mixing them improves the model’s overall agentic performance. We present ablation results on this effect in Section 5.

CUA Training Strategies Depending on computing budget, dataset availability, and the target model—e.g., building a specialized computer-use agent or a general-purpose VLM with agentic capabilities—different training strategies may be adopted. Following Aguvis [48], which proposes a two-stage curriculum training (Stage 1 for grounding and Stage 2 for planning and reasoning), we further explore three strategies:

1. **Stage 2 only:** When training resources are limited and the focus is on computer-use agent data, we aim to adapt a general open-source VLM into a specialized CUA. To preserve general instruction-following ability, we use a training mix of 70% CUA data (with a planning-to-grounding ratio of

4:1) and 30% general SFT data. We fine-tune Qwen2-VL with 30B tokens and Kimi-VL-A3B with 20B tokens. Both models exhibit strong improvements on CUA tasks (see Table 10).

2. **Stage 1 + Stage 2:** With more resources and diverse data, a staged approach yields better performance. In Stage 1, we enhance grounding and understanding using grounding trajectories, tutorial-style demonstrations, state-transition caption data, general vision language tasks, and general text SFT data. We trained Qwen2.5-VL-32B on this mixture for 40B tokens. In Stage 2, we shift focus to CUA planning, using 45% planning, 20% grounding, and the rest general data. This results in OPENCUA-32B, which achieves substantial gains in both grounding and planning benchmarks (Table 3, Figure 7).
3. **Joint training:** To build a general-purpose VLM with strong CUA abilities, we adopt joint training across domains with balanced data mixing. Given the complexity of multi-image trajectory data, we train these samples for 3 epochs to ensure sufficient learning. Based on Qwen2.5-VL-7B, we train a model on 200B tokens budget, maintaining a data ratio of 20% planning, 20% grounding, and 60% general. The resulting model, OPENCUA-7B, achieves state-of-the-art performance among 7B-scale open-source CUAs, achieving 27.3% success rate on OSWorld Online Evaluation Platform.

4 Experiments

4.1 Experimental Setup

Models Our experiments are conducted on multiple open-sourced vision-language models: Kimi-VL-A3B [33], Qwen2-VL-7B-Instruct [36], Qwen2.5-VL-7B-Instruct [6], and Qwen2.5-VL-32B-Instruct [6]³. Kimi-VL-A3B adopts a Mixture-of-Experts (MoE) architecture with 16B total parameters and 3B active during training and inference. It demonstrates moderate capability as a computer-use agent, including grounding and planning. Qwen2-VL and Qwen2.5-VL are general-purpose vision-language models, with Qwen2.5-VL exhibiting enhanced digital agent capabilities and expertise in high-resolution understanding. We conduct supervised fine-tuning (SFT) on these models and obtain our OpenCUA model variants: OPENCUA-A3B, OPENCUA-QWEN2-7B, OPENCUA-7B, and OPENCUA-32B.

Evaluation We evaluated our models on online evaluation benchmarks, offline agent evaluation benchmark, and GUI grounding benchmarks.

1. **Online agent evaluation:** (1) **OSWorld-Verified:** OSWorld [42] originally curated 369 human-crafted tasks covering a wide range of applications, each accompanied by its own environment setup and evaluation script. The OSWorld team has now verified every task, fixing those that were infeasible to test because of outdated dependencies, evaluation errors, or ambiguous instructions, and has released the improved benchmark as OSWorld-Verified [45]⁴. Our results are obtained through the public evaluation by the OSWorld Team on AWS infrastructure. The results are reported in Table 3. (2) **WindowsAgentArena (WAA)** [7] contains 154 Windows-centric tasks, spanning native Windows applications as well as several open-source programs also featured in OSWorld. It can reflect the agent’s online performance on Windows system.

Evaluation setting: The resolution of the systems is 1920×1080 . 4 clock tasks in WAA are dropped due to API and system image limitation. We adopt the L2 CoT format (Thought + Action) for all models, following results of our ablation in Section 5. Temperature is set to 0 for deterministic decoding during evaluation. The results in OSWorld-Verified are the average of 3 runs.

2. **Offline agent evaluation:** AGENTNETBENCH includes 100 representative held-out tasks covering diverse domains on Windows and macOS. The introduction and details of AGENTNETBENCH are in Appendix B. We also validated its correlation with online benchmark results in Figure 14.

³To align with the training infrastructure of the Kimi Team, we adopt the same chat template and tokenizer as Kimi-VL-A3B. M-RoPE in Qwen models is not implemented; we use 1D RoPE [32] instead.

⁴OSWorld-Verified Leaderboard: <https://os-world.github.io/> OSWorld-Verified blog: <https://xlang.ai/blog/osworld-verified>

3. **GUI grounding evaluation:** We evaluate our model’s GUI grounding ability, the ability to map natural language instructions to specific actions within graphical user interfaces on three benchmarks: OSWorld-G [44], Screenspot-V2 [40] and Screenspot-Pro [21]. OSWorld-G has 564 samples that systematically cover text matching, element recognition, layout understanding and fine-grained manipulation, with annotations for the element types required to solve each task. Screenspot-V2 includes screenshots from three platforms: mobile, desktop, and web. Screenspot-Pro focuses on high-resolution desktop environments, especially in professional settings.

Training settings. All models are trained on the Kimi Team’s infrastructure with the Megatron framework and DeepSpeed (ZeRO-3). We employ three training strategies:

1. **Stage-2 only.** OPENCUA-QWEN2-7B and OPENCUA-A3B share a configuration of sequence length 32,768, learning-rate 2×10^{-5} , weight-decay 0.1, and global batch size 384 (512 in ablations) on $96 \times \text{A100 GPUs}$. They are trained on 18k Win&macOS + 10k Ubuntu trajectories. OPENCUA-QWEN2-7B runs for 3,400 steps (about 45 h) after a 400-step grounding warm-up; OPENCUA-A3B runs for 2,000 steps (about 10 h).
2. **Stage-1 + Stage-2.** OPENCUA-32B is first pretrained on 35B tokens of general text, vision, and grounding data (batch 3,584, LR 3×10^{-5} , $224 \times \text{A100}$). We take the step-1200 checkpoint. Stage-2 then continues for 60B tokens on trajectory + general + grounding data (batch 512, LR 2.5×10^{-5} , $128 \times \text{A100}$) using 18k Win&macOS + 20k Ubuntu trajectories. The final model corresponds to step 4,700.
3. **Joint training.** OPENCUA-7B is trained end-to-end on the full data mixture for 200B tokens (18k Win&macOS + 20k Ubuntu trajectories) with batch 512, peak LR 2.5×10^{-5} (min LR 3×10^{-6}), decay tokens 200B, on $128 \times \text{A100}$ for eight days. The best checkpoint is at step 14,600.

4.2 Main Results

Table 3: **OSWorld-Verified results.** OPENCUA-32B achieves the best performance among all open-source models with an average success rate of **34.8%**, outperforming prior baselines by large margins. It also closes the gap to proprietary agents, even surpassing OpenAI CUA. This demonstrates the scalability and strength of our OpenCUA training pipeline. (The OPENCUA-7B and OPENCUA-32B are evaluated 3 times independently [Tao: then why not add Confidence intervals in the table? to directly show this](#). The details of each run and the Pass@3 evaluation results are shown in Appendix Table 10)

Model	15 Steps	50 Steps	100 Steps
Proprietary			
OpenAI CUA [28]	26.0	31.3	31.4
Seed1.5-VL [17]	27.9	-	34.1
Claude 3.7 Sonnet [5]	27.1	35.8	35.9
Claude 4 Sonnet [4]	<u>31.2</u>	<u>43.9</u>	<u>41.5</u>
Open-Source			
Qwen2.5-VL-32B-Instruct [6]	3.0	-	3.9
Qwen2.5-VL-72B-Instruct [6]	4.4	-	5.0
Kimi-VL-A3B [33]	9.7	-	10.3
UI-TARS-72B-DPO [29]	24.0	25.8	27.1
UI-TARS-1.5-7B [29]	24.5	27.3	27.4
OpenCUA-7B (Ours)	24.3	27.9	26.6
OpenCUA-32B (Ours)	29.7	34.1	34.8

Online agent evaluation. Table 3 lists success rates of end-to-end agents for 15-, 50- and 100-step budgets on **OSWorld-Verified**. The details of each run, Pass@3 evaluation success rate, and the results of OPENCUA-A3B and OPENCUA-7B are demonstrated in Appendix Table 10. Besides quantitative metrics, we also provide a qualitative case study in Appendix F, which shows that our agent can recognize and correct earlier mistakes during long-horizon execution, ultimately enabling more reliable task completion.

1. **Proprietary models still lead, but the gap is closing.** Claude 4 Sonnet achieves 43.9 % at 100 steps, followed by Claude 3.7 Sonnet at 35.9 %. Our OPENCUA-32B reaches 34.8 % in 100 steps, establishing a new state-of-the-art among open-source systems, surpassing the GPT-4o-based OpenAI CUA (31.4 %) and further closing the gap to Claude models.
2. **OpenCUA method applies to models with different architectures and sizes.** We apply our method on 4 model architectures, including Kimi-VL-A3B [33], Qwen2-VL-7B [36], Qwen2.5-VL-7B [6] and Qwen2.5-VL-32B [6], covering MoE and dense structures and activation parameter number from 3B to 32B. As the results in Appendix 10, all the models’ performance improves by a large margin upon the base models. In confirming that our data pipeline and training recipe scale effectively with model capacity. Furthermore, their performance scales with the model size -OPENCUA-32B consistently outperforms OPENCUA-7B under every step budget.
3. **Effect of step limit.** Most agents profit from a larger step budget from 15 to 50 steps, yet the gain from 50 to 100 steps is often smaller than that from 15 to 50 steps.
 - Claude 4 Sonnet: +12.7 % from 15 $\rightarrow$ 50 steps, then -2.4 % from 50 $\rightarrow$ 100.
 - OPENCUA-32B: +4.4 % (29.7 $\rightarrow$ 34.1) from 15 $\rightarrow$ 50 steps, and +0.7 % (34.1 $\rightarrow$ 34.8) from 50 $\rightarrow$ 100 steps.
 - Base models including Qwen2.5-VL and Kimi-VL: marginal gains (< 1 %), reflecting limited step scale capability of base models.

Even top-tier agents (e.g., Claude 3.5 Sonnet) still struggle to benefit significantly on larger step budgets for truly long-horizon tasks. The modest gain when moving from a 50-step to a 100-step limit stems from two main factors: (i) most tasks need more than 15 but fewer than 50 GUI actions, so the extra head-room often goes unused; (ii) current models are still not good at recognising their own mistakes, recovering from errors, and deciding when to stop—hallucinations and repetitive loops frequently waste the additional steps. We did a detailed error study in Appendix E.
4. **OpenCUA models achieve markedly higher Pass@n scores.** As reported in Appendix 10, the Pass@3 success rate of OPENCUA-32B on OSWORLD-Verified jumps from 34.2%(Pass@1) to **45.6%**. This large margin suggests ample headroom for future post-training, reranking or multi-agent methods. Additional analyses of test-time upper bounds and robustness are provided in Section 5.

These observations validate the effectiveness of our pipeline and highlight the remaining headroom for open-source agents to close the gap with larger proprietary models.

Offline benchmark evaluation. AGENTNETBENCH is constructed from representative tasks in the OPENCUA dataset. It is a held-out testset including 100 task trajectories on Windows and macOS. To account for domain similarity, we group models into *Zero-shot* systems—those not trained on OPENCUA (Qwen2.5-VL-7B/32B/72B, Aguis-7B, and OpenAI CUA (GPT-4o))—and *Fine-tuned* models, namely our OPENCUA-7B and OPENCUA-32B. As summarized in Table 4:

1. **Scaling with model size.** In the zero-shot group, performance scales with model size; the specialist Aguis-7B surpasses the general-purpose Qwen2.5-VL-7B.
2. **OpenAI CUA generalize well on unseen tasks.** OpenAI CUA outperforms all open-source zero-shot models and approaches fine-tuned agents, particularly excelling in terminate-state detection and content-based actions.
3. **Offline benchmark is consistent with the ranking on online benchmark.** The overall ordering matches the online leaderboard in Table 3: OPENCUA-32B $>$ OpenAI CUA $>$ OPENCUA-7B $>$ Qwen2.5-VL models.
4. **Coordinate-action performance reflects model grounding performance.** OPENCUA-32B exceeds OPENCUA-7B on coordinate-based actions, aligned the grounding results in Figure 7.

GUI grounding evaluation. Figure 7 and Table 3 reveal several key findings:

1. **OPENCUA-32B ranks the first.** OPENCUA-32B is the best-performing agentic model on all mainstream GUI-grounding benchmarks. Its advantage stems from (i) the substantially larger grounding corpus used in Stage-1 training and (ii) its larger 32B parameter scale.
2. **Joint-training boosts OPENCUA-7B.** OPENCUA-7B also scores competitively because large-scale grounding data are injected during joint training.

Table 4: Computer-use agent performance on AGENTNETBENCH. Coord actions: *click*, *rightClick*, *doubleClick*, *moveTo*, *dragTo*, *scroll*; Content actions: *write*, *press*, *hotkey*; Function action: *terminate*.

Model	Coord. SR	Content SR	Func. SR	Avg. SR
Qwen2.5-VL-7B [6]	50.7	40.8	3.1	48.0
Aguvis-7B [48]	56.7	43.3	0.0	52.4
Qwen2.5-VL-32B [6]	66.6	47.2	41.5	64.8
Qwen2.5-VL-72B [6]	67.2	52.6	50.5	67.0
OpenAI CUA [28]	<u>71.7</u>	57.3	80.0	<u>73.1</u>
OpenCUA-7B (Ours)	75.4	46.4	53.6	71.0
OpenCUA-32B (Ours)	78.7	46.0	55.2	73.2

- Pixel-budget advantage of Qwen2.5-VL.** Qwen2.5-VL based models uses a higher max-pixels limit (12,845,056 vs. 829,440 for Qwen2-VL and Kimi-VL-A3B), yielding stronger results on high-resolution ScreenSpot-Pro. OPENCUA-32B achieves 55.3% on Screenspot-Pro, and OPENCUA-7B achieves 50.0%.
- Grounding alone is not enough.** Although Qwen2.5-VL-32B matches or even surpasses OPENCUA-QWEN2-7B and OPENCUA-A3B on OSWorld-G and ScreenSpot-V2, OpenCUA models achieve far higher success rates on the full OSWorld benchmark (OPENCUA-A3B 19.9% and OPENCUA-QWEN2-7B 23.0% shown in Appendix 10). This shows that solid grounding is necessary yet insufficient for realistic agentic tasks in the environment; high-level planning and reflective reasoning ultimately drive reliable task completion.
- Domain correlation.** Because OSWorld-G [44] is collected in an Ubuntu environment, its scores correlate more closely with the online OSWorld evaluation than with ScreenSpot-Pro and Screenspot-V2.

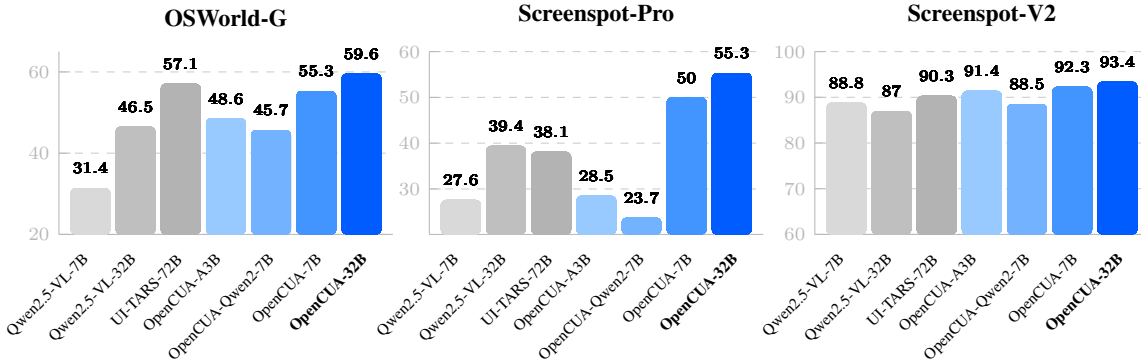


Figure 7: GUI Grounding performance of OpenCUA models compared with Qwen2.5-VL models and UI-TARS on three benchmarks.

Performance scaling with data scaling. We explore the effect of data scale on Qwen2-VL from three perspectives: cross-domain data, in-domain data, and out-of-domain data. We first investigate cross-domain data in Figure 1. Specifically, we compare three training settings: (1) 7K Ubuntu data, (2) 7K Ubuntu + 14K Win&Mac data, and (3) 10K Ubuntu + 17K Win&Mac data. On OSWorld, performance improves significantly from **9.8%** to **18.5%**, despite the added Win&Mac data coming from a different platform. This indicates that even out-of-domain data can substantially enhance generalization and reasoning ability, rather than causing negative transfer. To further study the impact of in-domain and out-of-domain data scale, we randomly sampled 3K, 7K, 10K trajectories from Ubuntu data and 3K, 7K 14K from Win&Mac.

As shown in Figure 8, performance scales consistently across all benchmarks with both in-domain and out-of-domain data. When increasing the Ubuntu data from 3K to 10K, the average performance improves by 72%. Scaling the Win/Mac data from 3K to 14K yields a 125% improvement on average. These results demonstrate a strong positive correlation between data quantity and agent performance, highlighting the importance of large-scale, diverse CUA data for model generalization.

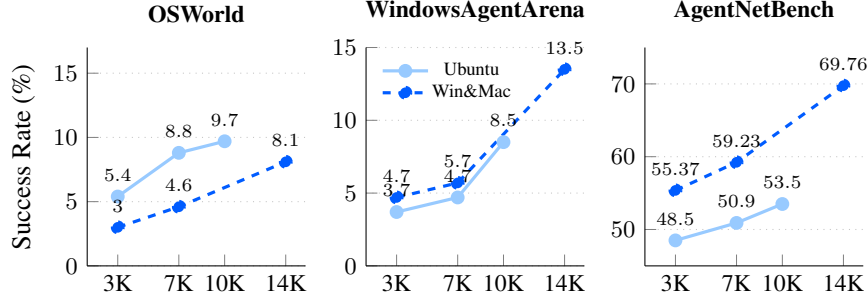


Figure 8: Scaling curves on three benchmarks as data volume from various OS domains increases.

5 Analysis

Model performance upperbound analysis by scaling test-time compute

We further explore our model’s performance upperbound by doing Pass@ n evaluation on OSWorld. We set the temperature to 0.1 and evaluate OPENCUA-QWEN2-7B for 16 times on the budget of 15, 30 and 50 steps and calculated the pass@1,4,8,16 success rate. In Table 9, we find: (1) There is a significant performance gap of our model between Pass@1 and Pass@16. On 15 step, the success rate increases from 16.9 to 34.6 (+104%), while on 50 step, the increacement is even large from 18.4 to 39.2 (+113%). (2) With larger n , the performance gains from increasing the step budget become more significant. (3) Online benchmarks have a large variance. To study model robustness, we did Pass@ n evaluation when temperature is 0 in Figure 5 and find higher temperature leads to higher Pass@ n performance but lower Pass@1 performance.

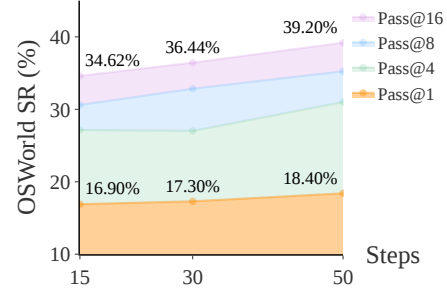


Figure 9: OSWorld Pass@N performance curves of OPENCUA-QWEN2-7B, temperature=0.1

We then investigated in the results and find variance comes from these factors:

1. The agent chooses different solutions in different runs. For example, in the task “Re-open the last closed browser tab,” the agent sometimes uses Ctrl + Shift + T (only one step) and other times navigates through the history menu (needs many steps). Agents may fail on harder solutions.
2. Minor omissions or extras. In Chrome or VS Code settings, forgetting to click “Save” (or performing an additional stray click) converts a correct solution into failure.
3. Environment dynamics: Occasional CAPTCHA dialogs, machine variability, and network latency can change the interaction sequence and lead to inconsistent outcomes.

Agent model is not robust: small variance in the environment affects the task result.

As illustrated in Figure 10, OPENCUA-QWEN2-7B’s OSWorld performance (Pass@N) under temperature=0 exhibits significant outcome divergence despite nearly identical initial states—with only minor variations (e.g., system date). The curves for Pass@16 (38.60% SR at 50 steps) and Pass@1 (20.10% SR) demonstrate a >18% absolute gap, highlighting how minimal initial perturbations propagate into starkly different trajectories. This underscores the model’s sensitivity to initial conditions even in deterministic (temp=0) settings, suggesting that seemingly trivial factors (e.g., temporal context) may critically influence multi-step reasoning.

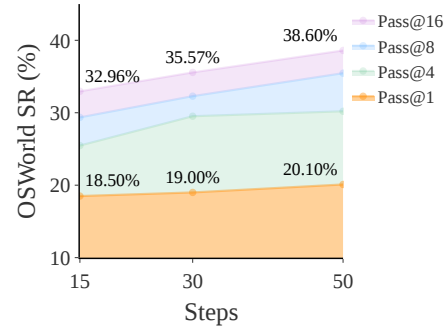


Figure 10: OSWorld Pass@N performance curves of OPENCUA-QWEN2-7B, temperature=0

Table 5: Ablation results on OSWorld for different Chain-of-Thought (CoT) settings.

Ablation	CoT Variant	SR (%)
CoT Mixture	L2	13.1
	Mixture-CoT	18.5
Reflective Long CoT	Short-CoT	11.5
	Advanced-CoT	15.3
Test-time Reasoning Format	L1	16.9
	L2	18.5
	L3	17.6



Figure 12: General text data ablation.

Cross-platform training improves generalization, even with domain differences.

As shown in Figure 8, there is a consistent performance gap between models trained on different domains. Models trained on Ubuntu data perform better on OSWorld, while those trained on Windows/macOS data perform better on WindowsAgentArena and AGENTNETBENCH. This domain gap reflects the underlying differences in GUI layouts, system styles, and application behavior across platforms. OSWorld primarily focuses on applications and websites aligned with Ubuntu environments, whereas WindowsAgentArena contains several OSWorld Windows-specific applications. Interestingly, the performance gap between training on Win&Mac data versus Ubuntu data is narrower on WAA than on OSWorld, suggesting that application-level knowledge can partially transfer across operating systems, even if interface styles differ.

L2 reasoning format achieves the best inference performance. Note that we trained the models with mixed reasoning format (L1, L2, L3, see Section 3.1). In this part, we explore which type of thinking format is the best at inference. We test OPENCUA-QWEN2-7B and OPENCUA-A3B on OSWorld in 15 steps. As in Table 5, using the L2 format, the performance is higher than L1 and L3. This result is actually different from the conclusion from previous work [48, 29] that L1 outperforms L2. We think this is because our L2 CoT has higher quality (e.g., planning and reflection), which can help the model make better decisions. On the other hand, L3 underperforms L2. By case study, we find that when model describes the information in the screenshot, there tend to be many elements irrelevant to the task or the next action, which may mislead the model. In summary, our results show that the right choice of high-quality, non-redundant reasoning can leverage VLM’s language reasoning capability to improve the agent performance.

Using a moderate number of visual history images and concise textual history yields the best trade-off between performance and efficiency.

We ablate history representation from both visual and textual perspectives. For vision, we vary the number of history screenshots (1, 3, 5) and fine-tune Qwen2-VL-7B on 7K trajectories. As the OSWorld results shown in Figure 11, using multiple screenshots substantially improves performance over single-image inputs, as GUI agents rely entirely on vision for observing state changes. However, increasing from 3 to 5 images yields marginal gains while incurring 3K more context tokens and delayed convergence, suggesting diminishing returns.

On the textual side, we compare L1 and L2 history under the same 3-image setting. In Figure 11, L2 history offers no benefit and may introduce hallucinations that distract attention, while also reducing training efficiency. Hence, we adopt L1 CoT + 3 images as the default setting.

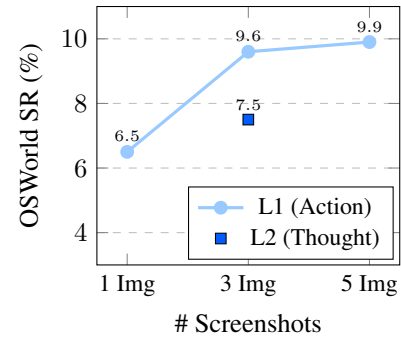


Figure 11: Effect of history representation: L1 (Action) benefits from more screenshots, while L2 (Thought) at 3 images lags behind.

Training with a mixture of CoT formats outperforms using only L2 reasoning. Since our best performance is from L2 CoT inference, and L3 and L1 is lower than L2, we did an ablation of

only training the L2 data instead of the mixture of L1, L2, and L3. We use the same recipe as our OPENCUA-QWEN2-7B, but only replace the mixture CoT data with L2 data. As the OSWorld result in Table 5, the model trained on L2 data using the same steps as OPENCUA-QWEN2-7B, but the performance drops to 13.1, which is aligned with the conclusion of Aguis [48].

General-domain text data provides a positive effect to agent performance. As we mentioned in Section 3.3, we used 35% general text data in our main experiment, so we also use the same agent data without the text data to fine-tune Qwen2-VL-7B with grounding warm-up stage for 2400 steps (approximately the same amount of agent data tokens) to ablate its influence. According to Figure 12, the general text data slightly improves model’s agentic performance. Therefore, adding text data from totally different general domains doesn’t impair the agent model’s performance, on the contrary, helps improve the performance. We think the reason is that the general text data may help agent model’s generalization and instruction understanding.

Reflective long CoT significantly boosts performance by improving error correction. To understand the effect of reflective long CoT (Section 3.1), we do an ablation study on Qwen2-VL-7B with 14K Win&Mac and 3K Ubuntu trajectories. Without reflective long CoT, the CoT reduces to that used by Aguis [48]. In Figure 6, we see that reflective long CoT improves the performance from 11.5 to 15.3. Since the reflective reasoning focuses on error correction, we conjecture that the improvement comes from improved self-correction capability.

6 Related Work

CUA benchmarks and datasets Autonomous computer-use agents are now judged primarily through *execution-level* benchmarks that embed the agent inside genuine software environments rather than synthetic simulators. On the desktop side, suites such as [43, 7, 53] orchestrate multi-step workflows that span office productivity, source-code editing, file management, and operating-system configuration across Linux, Windows, and macOS. For the web domain, campaigns including [54, 19, 12, 49, 13] deploy agents on self-hosted or live sites with dynamic content, long navigation chains, and non-trivial information-retrieval subtasks. To support training at the necessary scale, several high-volume data pipelines have appeared: tutorial-to-trajectory conversion for browser tasks [46], cross-device grounding and action logs [18, 10, 16], plus our own collection of 27 K desktop demonstrations that pair screenshots, low-level mouse/keyboard events, and reflective chain-of-thought annotations.

CUA frameworks and models Approaches to building computer-use agents can be grouped into three broad categories. First, text-based language models operate on structured GUI metadata—such as DOM trees or accessibility labels—and issue symbolic commands; representative work ranges from early page-centric agents [25] to more recent language-only planners that still eschew raw pixels [47]. Second, vision-centric agents integrate screen imagery. Some focus on grounding—learning to associate natural-language references with bounding boxes or coordinate clicks [16, 40, 44]—while others pursue end-to-end policies that translate full screenshots directly into action sequences [48, 29, 28, 3, 33, 17]. Third, agent frameworks (in-context learning/modular/prompting based agents) wrap large language models with additional components—specialised vision encoders, hierarchical or search-based planners, episodic memory, and tool APIs—to tackle long-horizon tasks requiring perception, reasoning, and control [55, 1, 2, 20].

7 Conclusion

We presented OPENCUA, a comprehensive open-source framework addressing critical gaps in computer-use agent development. By offering annotation infrastructure, data processing pipelines, diverse datasets, effective training recipes, and efficient evaluation benchmarks, we establish essential foundations for CUA research. Our models demonstrate strong performance across benchmarks while exhibiting clear data scaling laws and cross-domain generalization capabilities. By releasing all components—tools, datasets, code, and models—we aim to accelerate transparent CUA research, enabling the community to systematically investigate these agents’ capabilities, limitations, and risks as they increasingly mediate our digital interactions and execute consequential decisions on our behalf.

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References

- [1] Saaket Agashe, Jiuzhou Han, Shuyu Gan, Jiachen Yang, Ang Li, and Xin Eric Wang. Agent s: An open agentic framework that uses computers like a human. *arXiv preprint arXiv:2410.08164*, 2024.
- [2] Saaket Agashe, Kyle Wong, Vincent Tu, Jiachen Yang, Ang Li, and Xin Eric Wang. Agent s2: A compositional generalist-specialist framework for computer use agents. *arXiv preprint arXiv:2504.00906*, 2025.
- [3] Anthropic. Claude computer use. <https://www.anthropic.com/news/3-5-models-and-computer-use>, 2024. Accessed: 2025-05-03.
- [4] Anthropic. Introducing claude 4, 5 2025. URL <https://www.anthropic.com/news/claude-4>. Accessed: 2025-07-28.
- [5] Anthropic. Claude’s extended thinking. <https://www.anthropic.com/research/visible-extended-thinking>, 2025. Accessed: 2025-05-03.
- [6] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yanzhi Zhu, Ming-Hsuan Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report. *CoRR*, abs/2502.13923, 2025.
- [7] Rogerio Bonatti, Dan Zhao, Francesco Bonacci, Dillon Dupont, Sara Abdali, Yinheng Li, Yadong Lu, Justin Wagle, Kazuhito Koishida, Arthur Buckner, Lawrence Jang, and Zack Hui. Windows agent arena: Evaluating multi-modal os agents at scale, 2024. URL <https://arxiv.org/abs/2409.08264>.
- [8] Yuxiang Chai, Siyuan Huang, Yazhe Niu, Han Xiao, Liang Liu, Dingyu Zhang, Shuai Ren, and Hongsheng Li. Amex: Android multi-annotation expo dataset for mobile gui agents. *arXiv preprint arXiv:2407.17490*, 2024.
- [9] Wentong Chen, Junbo Cui, Jinyi Hu, Yujia Qin, Junjie Fang, Yue Zhao, Chongyi Wang, Jun Liu, Guirong Chen, Yupeng Huo, et al. Guicourse: From general vision language models to versatile gui agents. *arXiv preprint arXiv:2406.11317*, 2024.
- [10] Kanzhi Cheng, Qiushi Sun, Yougang Chu, Fangzhi Xu, Yantao Li, Jianbing Zhang, and Zhiyong Wu. SeeClick: Harnessing gui grounding for advanced visual gui agents. *arXiv preprint arXiv:2401.10935*, 2024.
- [11] Edoardo DeBenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. Agentdojo: A dynamic environment to evaluate prompt injection attacks and defenses for llm agents, 2024. URL <https://arxiv.org/abs/2406.13352>.
- [12] Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Sam Stevens, Boshi Wang, Huan Sun, and Yu Su. Mind2web: Towards a generalist agent for the web. *Advances in Neural Information Processing Systems*, 36:28091–28114, 2023.
- [13] Alexandre Drouin, Maxime Gasse, Massimo Caccia, Issam H. Laradji, Manuel Del Verme, Tom Marty, David Vázquez, Nicolas Chapados, and Alexandre Lacoste. Workarena: How capable are web agents at solving common knowledge work tasks? In *Forty-first International Conference on Machine Learning*, 2024. URL <https://openreview.net/forum?id=BRfqYrikdo>.
- [14] Boyu Gou, Ruohan Wang, Boyuan Zheng, Yanan Xie, Cheng Chang, Yiheng Shu, Huan Sun, and Yu Su. Navigating the digital world as humans do: Universal visual grounding for gui agents. *arXiv preprint arXiv:2410.05243*, 2024.
- [15] Boyu Gou, Ruohan Wang, Boyuan Zheng, Yanan Xie, Cheng Chang, Yiheng Shu, Huan Sun, and Yu Su. Navigating the digital world as humans do: Universal visual grounding for GUI agents. *CoRR*, abs/2410.05243, 2024. URL <https://doi.org/10.48550/arXiv.2410.05243>.

- [16] Boyu Gou, Ruohan Wang, Boyuan Zheng, Yanan Xie, Cheng Chang, Yiheng Shu, Huan Sun, and Yu Su. Navigating the digital world as humans do: Universal visual grounding for GUI agents. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [17] Dong Guo, Faming Wu, Feida Zhu, Fuxing Leng, Guang Shi, Haobin Chen, Haoqi Fan, Jian Wang, Jianyu Jiang, Jiawei Wang, et al. Seed1. 5-vl technical report. *arXiv preprint arXiv:2505.07062*, 2025.
- [18] Raghav Kapoor, Yash Parag Butala, Melisa Russak, Jing Yu Koh, Kiran Kamble, Waseem AlShikh, and Ruslan Salakhutdinov. Omniaact: A dataset and benchmark for enabling multimodal generalist autonomous agents for desktop and web. In *European Conference on Computer Vision*, pages 161–178. Springer, 2024.
- [19] Jing Yu Koh, Robert Lo, Lawrence Jang, Vikram Duvvur, Ming Chong Lim, Po-Yu Huang, Graham Neubig, Shuyan Zhou, Russ Salakhutdinov, and Daniel Fried. Visualwebarena: Evaluating multimodal agents on realistic visual web tasks. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.
- [20] Guohao Li, Hasan Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. Camel: Communicative agents for "mind" exploration of large language model society. *Advances in Neural Information Processing Systems*, 36:51991–52008, 2023.
- [21] Kaixin Li, Ziyang Meng, Hongzhan Lin, Ziyang Luo, Yuchen Tian, Jing Ma, Zhiyong Huang, and Tat-Seng Chua. Screenspot-pro: Gui grounding for professional high-resolution computer use. *arXiv preprint arXiv:2504.07981*, 2025.
- [22] Wei Li, William Bishop, Alice Li, Chris Rawles, Folawiyo Campbell-Ajala, Divya Tyamagundlu, and Oriana Riva. On the effects of data scale on computer control agents, 2024. URL <https://arxiv.org/abs/2406.03679>.
- [23] Kevin Qinghong Lin, Linjie Li, Difei Gao, Zhengyuan Yang, Shiwei Wu, Zechen Bai, Weixian Lei, Lijuan Wang, and Mike Zheng Shou. Showui: One vision-language-action model for gui visual agent. *arXiv preprint arXiv:2411.17465*, 2024.
- [24] Quanfeng Lu, Wenqi Shao, Zitao Liu, Fanqing Meng, Boxuan Li, Botong Chen, Siyuan Huang, Kaipeng Zhang, Yu Qiao, and Ping Luo. Gui odyssey: A comprehensive dataset for cross-app gui navigation on mobile devices. *arXiv preprint arXiv:2406.08451*, 2024.
- [25] Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, et al. Webgpt: Browser-assisted question-answering with human feedback. *ArXiv preprint*, 2021. URL <https://arxiv.org/abs/2112.09332>.
- [26] OBS Project. Obs studio. URL <https://github.com/obsproject/obs-studio>. Accessed 2025-08-04.
- [27] OpenAdaptAI. Openadapt: Ai-first process automation with large multimodal models (lmms). URL <https://github.com/OpenAdaptAI/OpenAdapt>. Accessed 2025-08-04.
- [28] OpenAI. Operator, 2025. URL <https://openai.com/research/operator>.
- [29] Yujia Qin, Yining Ye, Junjie Fang, Haoming Wang, Shihao Liang, Shizuo Tian, Junda Zhang, Jiahao Li, Yunxin Li, Shijue Huang, Wanjuan Zhong, Kuanye Li, Jiale Yang, Yu Miao, Woyu Lin, Longxiang Liu, Xu Jiang, Qianli Ma, Jingyu Li, Xiaojun Xiao, Kai Cai, Chuang Li, Yaowei Zheng, Chaolin Jin, Chen Li, Xiao Zhou, Minchao Wang, Haoli Chen, Zhaojian Li, Haihua Yang, Haifeng Liu, Feng Lin, Tao Peng, Xin Liu, and Guang Shi. Ui-tars: Pioneering automated gui interaction with native agents, 2025. URL <https://arxiv.org/abs/2501.12326>.
- [30] Christopher Rawles, Alice Li, Daniel Rodriguez, Oriana Riva, and Timothy Lillicrap. Androidinthewild: A large-scale dataset for android device control. *Advances in Neural Information Processing Systems*, 36:59708–59728, 2023.

- [31] Yangjun Ruan, Honghua Dong, Andrew Wang, Silviu Pitis, Yongchao Zhou, Jimmy Ba, Yann Dubois, Chris J. Maddison, and Tatsunori Hashimoto. Identifying the risks of lm agents with an lm-emulated sandbox. *ArXiv*, abs/2309.15817, 2023. URL <https://api.semanticscholar.org/CorpusID:262944419>.
- [32] Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 2024.
- [33] Kimi Team, Angang Du, Bohong Yin, Bowei Xing, Bowen Qu, Bowen Wang, Cheng Chen, Chenlin Zhang, Chenzhuang Du, Chu Wei, et al. Kimi-vl technical report. *arXiv preprint arXiv:2504.07491*, 2025.
- [34] TheDuckAI. Ducktrack. URL <https://github.com/TheDuckAI/DuckTrack>. Accessed 2025-08-04.
- [35] Bowen Wang, Xinyuan Wang, Jiaqi Deng, Tianbao Xie, Ryan Li, Yanzhe Zhang, Gavin Li, Toh Jing Hua, Ion Stoica, Wei-Lin Chiang, Diyi Yang, Yu Su, Yi Zhang, Zhiguo Wang, Victor Zhong, and Tao Yu. Computer agent arena: Compare test computer use agents on crowdsourced real-world tasks, 2025.
- [36] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Yang Fan, Kai Dang, Mengfei Du, Xuancheng Ren, Rui Men, Dayiheng Liu, Chang Zhou, Jingren Zhou, and Junyang Lin. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *ArXiv preprint*, 2024. URL <https://arxiv.org/abs/2409.12191>.
- [37] Michael Wornow, Avani Narayan, Ben Viggiano, Ishan S. Khare, Tathagat Verma, Tibor Thompson, Miguel Angel Fuentes Hernandez, Sudharsan Sundar, Chloe Trujillo, Krrish Chawla, Rongfei Lu, Justin Shen, Divya Nagaraj, Joshua Martinez, Vardhan Agrawal, Althea Hudson, Nigam H. Shah, and Christopher Re. Wonderbread: A benchmark for evaluating multimodal foundation models on business process management tasks, 2024. URL <https://arxiv.org/abs/2406.13264>.
- [38] Chen Henry Wu, Jing Yu Koh, Ruslan Salakhutdinov, Daniel Fried, and Aditi Raghunathan. Dissecting adversarial robustness of multimodal lm agents. In *International Conference on Learning Representations*, 2024. URL <https://api.semanticscholar.org/CorpusID:270562791>.
- [39] Penghao Wu and Saining Xie. V*: Guided visual search as a core mechanism in multimodal llms, 2023. URL <https://arxiv.org/abs/2312.14135>.
- [40] Zhiyong Wu, Zhenyu Wu, Fangzhi Xu, Yian Wang, Qiushi Sun, Chengyou Jia, Kanzhi Cheng, Zichen Ding, Liheng Chen, Paul Pu Liang, and Yu Qiao. OS-ATLAS: A foundation action model for generalist GUI agents. *CoRR*, abs/2410.23218, 2024. doi: 10.48550/ARXIV.2410.23218. URL <https://doi.org/10.48550/arXiv.2410.23218>.
- [41] Zhiyong Wu, Zhenyu Wu, Fangzhi Xu, Yian Wang, Qiushi Sun, Chengyou Jia, Kanzhi Cheng, Zichen Ding, Liheng Chen, Paul Pu Liang, and Yu Qiao. Os-atlas: A foundation action model for generalist gui agents, 2024. URL <https://arxiv.org/abs/2410.23218>.
- [42] Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, Yitao Liu, Yiheng Xu, Shuyan Zhou, Silvio Savarese, Caiming Xiong, Victor Zhong, and Tao Yu. Osworld: Benchmarking multimodal agents for open-ended tasks in real computer environments, 2024.
- [43] Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, et al. Osworld: Benchmarking multimodal agents for open-ended tasks in real computer environments. *ArXiv preprint*, 2024. URL <https://arxiv.org/abs/2404.07972>.
- [44] Tianbao Xie, Jiaqi Deng, Xiaochuan Li, Junlin Yang, Haoyuan Wu, Jixuan Chen, Wenjing Hu, Xinyuan Wang, Yuhui Xu, Zekun Wang, Yiheng Xu, Junli Wang, Doyen Sahoo, Tao Yu, and Caiming Xiong. Scaling computer-use grounding via user interface decomposition and synthesis, 2025. URL <https://arxiv.org/abs/2505.13227>.

- [45] Tianbao Xie, Mengqi Yuan, Danyang Zhang, Xinzhuang Xiong, Zhennan Shen, Zilong Zhou, Xinyuan Wang, Yanxu Chen, Jiaqi Deng, Junda Chen, Bowen Wang, Haoyuan Wu, Jixuan Chen, Junli Wang, Dunjie Lu, Hao Hu, and Tao Yu. Introducing osworld-verified. *xlang.ai*, July 2025. URL <https://xlang.ai/blog/osworld-verified>.
- [46] Yiheng Xu, Dunjie Lu, Zhennan Shen, Junli Wang, Zekun Wang, Yuchen Mao, Caiming Xiong, and Tao Yu. Agentrek: Agent trajectory synthesis via guiding replay with web tutorials. *arXiv preprint arXiv:2412.09605*, 2024.
- [47] Yiheng Xu, Hongjin Su, Chen Xing, Boyu Mi, Qian Liu, Weijia Shi, Binyuan Hui, Fan Zhou, Yitao Liu, Tianbao Xie, Zhoujun Cheng, Siheng Zhao, Lingpeng Kong, Bailin Wang, Caiming Xiong, and Tao Yu. Lemur: Harmonizing natural language and code for language agents. In *International Conference on Learning Representations*, 2024.
- [48] Yiheng Xu, Zekun Wang, Junli Wang, Dunjie Lu, Tianbao Xie, Amrita Saha, Doyen Sahoo, Tao Yu, and Caiming Xiong. Aguis: Unified pure vision agents for autonomous gui interaction. *arXiv preprint arXiv:2412.04454*, 2024.
- [49] Tianci Xue, Weijian Qi, Tianneng Shi, Chan Hee Song, Boyu Gou, Dawn Song, Huan Sun, and Yu Su. An illusion of progress? assessing the current state of web agents, 2025. URL <https://arxiv.org/abs/2504.01382>.
- [50] Zonghan Yang, Peng Li, Ming Yan, Ji Zhang, Fei Huang, and Yang Liu. React meets actre: When language agents enjoy training data autonomy. *arXiv preprint arXiv:2403.14589*, 2024.
- [51] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023.
- [52] Jiwen Zhang, Jihao Wu, Yihua Teng, Minghui Liao, Nuo Xu, Xiao Xiao, Zhongyu Wei, and Duyu Tang. Android in the zoo: Chain-of-action-thought for gui agents. *ArXiv preprint*, 2024. URL <https://arxiv.org/abs/2403.02713>.
- [53] Longtao Zheng, Zhiyuan Huang, Zhenghai Xue, Xinrun Wang, Bo An, and Shuicheng Yan. Agentstudio: A toolkit for building general virtual agents, 2025. URL <https://arxiv.org/abs/2403.17918>.
- [54] Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. Webarena: A realistic web environment for building autonomous agents. In *International Conference on Learning Representations*, 2024.
- [55] Wangchunshu Zhou, Yuchen Eleanor Jiang, Long Li, Jialong Wu, Tiannan Wang, Shi Qiu, Jintian Zhang, Jing Chen, Ruipu Wu, Shuai Wang, Shiding Zhu, Jiyu Chen, Wentao Zhang, Xiangru Tang, Ningyu Zhang, Huajun Chen, Peng Cui, and Mrinmaya Sachan. Agents: An open-source framework for autonomous language agents, 2023. URL <https://arxiv.org/abs/2309.07870>.

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A Limitations

The scalability of AGENTNET dataset is inherently limited by human annotation efforts. Although AGENTNET TOOL streamlines the data collection process, expanding the dataset beyond its current size would require additional human resources. Exploring alternative data sources or semi-automated annotation methods could help address this limitation. Additionally, though OPENCUA strives to collect authentic computer-use data from personal devices, our ethical requirement for explicit informed consent regarding data practices inevitably introduces selection bias. While our dataset maintains high diversity and authenticity, it necessarily excludes data from users who, upon understanding the potential risks, opt not to participate. This is a limitation we accept to uphold responsible data collection.

B AGENTNETBENCH

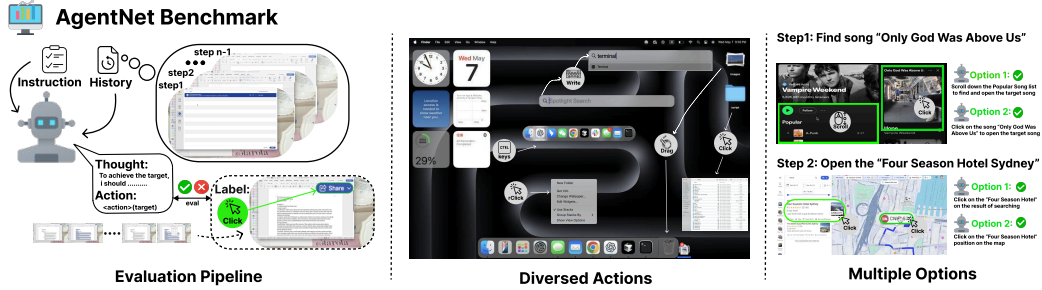


Figure 13: Illustration of the AgentNet Benchmark evaluation pipeline

There are several online benchmarks [42, 7] that evaluate agent performance in desktop environments. However, these online benchmarks typically require substantial computational resources for environment setup, making evaluations expensive, slow, and difficult to reproduce consistently through time due to their reliance on dynamic environments. Meanwhile, they only provide sparse, high-variance signals (i.e., trajectory-level accuracy). Another line of works, such as ComputerAgentArena [35], evaluate agent’s performance of open-ended tasks on live environments through human preference. To address the limitations of online evaluation benchmarks, we introduce an offline CUA evaluation benchmark, AGENTNETBENCH, comprising 100 representative tasks selected from the AGENTNET dataset. Tasks were strategically chosen from the center of sub-domain clusters (as detailed in Section C.1.1), ensuring diversity and representativeness across applications and websites on Windows and macOS platforms. Each task was manually reviewed to refine goals and remove redundant actions. Notably, we manually provide multiple valid action options at each step because of the inherent multiplicity of valid actions in computer-use tasks.

Benchmark statistics and evaluation dimensions The AGENTNETBENCH maintains a balanced domain distribution consisting of 38 Work tasks, 29 Daily tasks, 24 Professional tasks, and 9 System & Web Setup tasks. The tasks are split between two operating systems, with 61 tasks from Windows and 39 tasks from macOS. Screen resolutions are categorized into three levels (high, medium, and low) as detailed in Table 6 (note that, for practical purposes, all images in the benchmark are resized from their original resolutions). The distribution of actions within these tasks and additional benchmark statistics are also presented comprehensively in Table 6.

Multiple action choices for enhanced accuracy Previous offline benchmarks [30, 22] typically define a single ground-truth action at each step. This practice can negatively impact accuracy by disregarding alternative valid choices that an agent may reasonably make in real-world interactions. In contrast, in AGENTNETBENCH, we annotate multiple plausible action choices for each step to better reflect real-world decision-making variability.

Step success rate calculation and action matching criteria To calculate the Step Success Rate (Step SR), we evaluate the correctness of agent actions at each individual step using precise matching

Table 6: Comprehensive Statistics of AgentNetBench

Domain Distribution		Operating System Distribution	
Work	38	Windows	61
Daily	29	macOS (Darwin)	39
Professional	24		
System & Web Setup	9		
Resolution Distribution		Overall Statistics	
High	20	Total Tasks	100
Medium	33	Avg. Steps/Task	17.63
Low	47	Total Actions	2143
Action Distribution			
click	850 (67.0%)	doubleClick	19 (1.5%)
rightClick	17 (1.3%)	press	28 (2.2%)
dragTo	27 (2.1%)	write	137 (10.8%)
moveTo	45 (3.5%)	hotkey	30 (2.3%)
scroll	18 (1.4%)	terminate	100 (7.6%)

criteria tailored to different action types. For coordinate-based actions (*e.g.*, `click`, `doubleClick`, `moveTo`, `dragTo`, `rightClick`, and `hscroll`), we define bounding boxes around each action’s target location; the agent earns the step success point if its predicted coordinates fall within these bounding boxes. For content- or keyboard-based actions, such as `write`, we measure correctness by computing the edit distance between the predicted and ground-truth text; actions like `hotkey` and `press` require perfect matches of the specified key combinations. For the `scroll` action, correctness depends on two key criteria: the agent’s output coordinates must be within the designated bounding box, and the scrolling direction must exactly match the ground truth. Finally, the correctness of the `terminate` action depends on the agent appropriately terminating at precisely the correct step – neither prematurely nor delayed. Considering the distribution of actions (see Table 6), these fine-grained evaluation rules ensure accurate and fair evaluation of agent capabilities in diverse interaction scenarios.

AGENTNETBENCH strongly correlates with online benchmark performance The offline benchmark primarily assesses an agent’s decision-making capability by evaluating its first-choice accuracy at each task step. While agents can leverage self-reflection to recover from errors made in earlier steps, offline and online SRs should correlate under a low step budget. Figure 14 and Table 4 indeed demonstrate a clear positive correlation, specifically following a power-law relation between the online task success rate (under a 15 step budget) and the offline step success rate. Therefore, metrics obtained from our offline benchmark provide a reliable indicator of an agent’s foundational proficiency and its adaptability to realistic, resource-constrained online tasks.

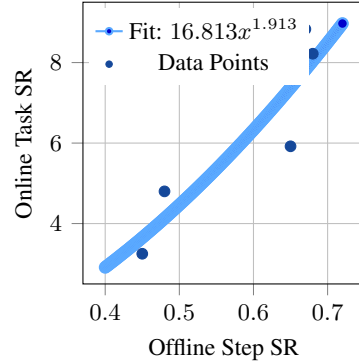


Figure 14: Offline vs. Online evaluation.

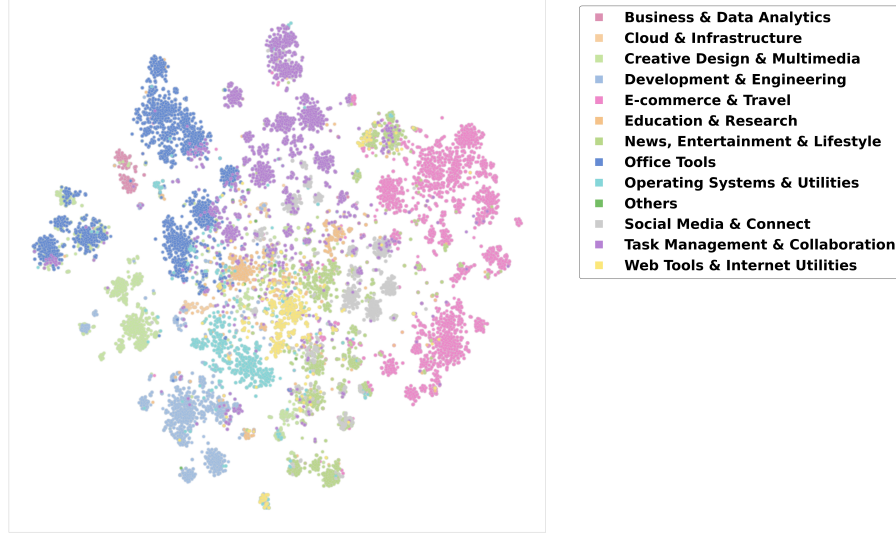


Figure 15: Clustering and t-SNE visualization of all task domains.

C Dataset Statistics, Annotation Details, and AgentNetTool Details

C.1 AGENTNET Statistics and Analysis

C.1.1 Diversity

Task Domains We categorize the collected tasks into 4 main domains and 11 subdomains based on their topics, involved applications and actions in the tasks (Figure 4). Table 7 lists representative applications for each domain. To label each task trajectory, we leveraged GPT-4o to complete the classification by representing each task using the task instruction and L1-level CoT. We manually examined 200 tasks randomly and the classification accuracy is over 96%. We then embedded the task trajectories using OpenAI’s `text-embedding-3-small` model and visualize them t-SNE visualization in Figure 15. Interestingly, the layout mirrors typical computer-usage patterns: for instance, Office Tools cluster near Business & Data-Analytics, while E-commerce & Travel sit close to Social-Media & Connect on the opposite side of the map. Finally, we chose 100 representative tasks around the cluster centroids to form our offline benchmark, AGENTNETBENCH.

Domain	App/Web
E-commerce & Travel	Amazon.com, Booking.com
News, Entertainment & Lifestyle	Spotify, Netflix
Social Media & Communication	WhatsApp, Instagram
Office Tools	Microsoft Office, Google Docs
Task Management & Collaboration	Zoom, Gmail, Slack
Creative Design & Multimedia	Photoshop
Development & Engineering	VSCode, PyCharm, Git
Knowledge Discovery & Research	Google Scholar, ResearchGate
Data Analysis, Business & Cloud	Tableau, Power BI, AWS
Web Tools & Internet Utilities	Chrome Extensions
Operating Systems & Utilities	Finder, Activity Monitor

Table 7: Example App/Web by Doman

Applications and Websites Applications and websites are tracked using the AgentNet Tool. Specifically, application names are captured by recording the process name when a user opens an application, while website URLs are recorded through our browser plugin. Discrepancies in process names across different operating systems and different versions are resolved using GPT, achieving an accuracy

of up to 83% with this combined method. Based on these results, we analyzed the distribution of the dataset across applications and websites. Web browsers account for a significant portion of the dataset, as nearly half of the data involves web applications. The results indicate that the dataset encompasses over 140 applications and 190 mainstream websites.

C.1.2 Complexity

Our collected tasks average 17.2 steps per task. We evaluate task complexity across five dimensions: multi-application/website usage, professional knowledge requirement, uncommon feature usage, repetitive simple subtasks, and logically coherent long sequences. Based on GPT-4o analysis, 30.6% tasks require multiple applications/websites, 12.9% involve professional knowledge, and 12.9% use uncommon features in Figure 17. Then we asked GPT to rate the complexity of tasks on a 1-10 scale, where 1 represents basic operations like file opening, and 10 indicates complex tasks requiring multiple steps, domain knowledge, or sophisticated reasoning. The complexity distribution is shown in Figure 16. It can be seen that most of the tasks have a medium or high level of complexity.

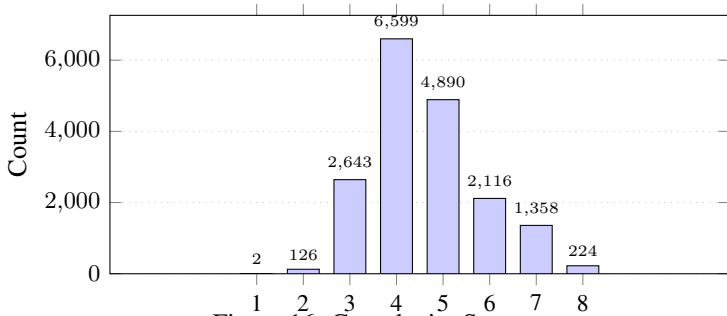


Figure 16: Complexity Score

C.1.3 Action distribution

Table 8 summarizes the action frequencies across 3 systems. We counted the action distribution of 41,428 computer-use tasks, including Ubuntu 23,797 tasks, Windows 12,431, and macOS 5,200. Click is the dominant action—more than 60% on three systems. Hardware and usage patterns drive the secondary behaviors: macOS trackpads lead to heavier vertical/horizontal scrolling and more hotkey use; Windows mouse workflows show higher proportions of right-click and middle-click; and Ubuntu’s keyboard-centric, terminal-oriented culture results in the greatest shares of text input.

Table 8: Action-type distribution (%) per operating system.

System	click	doubleClick	dragTo	hotkey	hscroll	middleClick	moveTo
Windows	60.03	1.99	4.35	1.52	0.00	0.01	8.95
macOS	57.22	2.20	3.48	2.48	0.46	0.00	9.02
Ubuntu	63.62	3.19	2.30	2.21	0.00	0.00	4.53
System	press	rightClick	scroll	terminate	tripleClick	write	
Windows	5.27	1.10	4.51	4.42	0.00	7.85	
macOS	6.39	1.01	5.52	4.40	0.08	7.73	
Ubuntu	3.78	2.40	2.06	5.89	0.22	9.81	

C.2 Annotation Details: Annotation Strategy, Annotator Source and Cost

Annotation Strategy We first surveyed the most popular websites and applications across a wide range of domains—entertainment, office tools, and more. We selected the 200 + most widely used ones. Tasks were not pre-assigned, because annotators’ familiarity varies (especially with professional tools). We set a limit number for the apps and allowed annotators to choose. Annotators could also brainstorm new tasks with provided relevant YouTube tutorials so they could explore and create additional tasks.

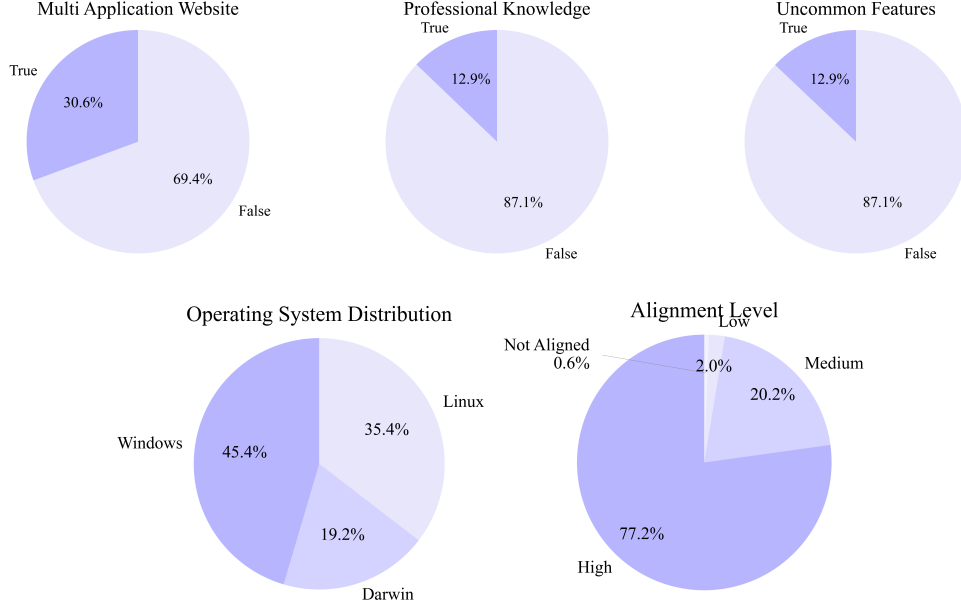


Figure 17: Distributions of data characteristics: presence of multi-application websites, inclusion of professional knowledge, presence of uncommon features, source operating systems and alignment levels.

Annotator Source We recruited annotators from four sources: internal students, external university students, annotation companies, and crowd-sourcing platform - Prolific. Table 9 shows the distribution of annotators and tasks. While annotators from Prolific and Company1 were native English speakers, others were native Chinese speakers. All annotators were required to document task goals in English and try to use English system settings, applications and websites to ensure broader applicability.

Annotator Source	Accepted Uploads	Annotator Count
Internal Students	4943	38
External Students	5168	135
Prolific	1218	294
Company1	2235	72
Company2	3556	51
Company3	1975	14
Company4	3000	30
Total	22095	634

Table 9: Annotation Source Statistics (before verification)

Annotation and CoT synthesis cost Annotating 22K tasks takes 6 months. All annotators are part-time. The total annotation cost was about USD 20,000. Annotation speed is roughly ten tasks per hour. The cost of synthesizing CoT costs USD 0.6 per task on average. The total cost of building this dataset is about USD 32,000.

C.3 AGENTNET TOOL: System Design and Key Features

C.3.1 Tool Features

1. **Action Reduction:** We use tools like pynput to capture users’ atomic actions. These atomic actions are then reduced to semantically meaningful actions, such as ‘click’, ‘key_press’, ‘key_release’,

‘type’, ‘drag’, ‘move’, and ‘scroll’. This reduction enables models to more effectively learn from human demonstrations and allows annotators and verifiers to understand trajectories more easily.

2. A11y Tree and HTML Processing: To obtain textual representations of observations, we implement efficient fetching and processing mechanisms for accessibility (a11y) trees and HTML. For the a11y tree, we apply pruning rules to select only essential element attributes, ensuring the fetching process for each tree takes minimal time. For HTML, we develop a browser plugin that automatically captures the HTML structure of websites.

3. Element Localization: To help users verify the correctness of their actions, we extract text associated with click locations. Using the a11y tree or HTML, we fetch the bounding box most likely clicked and extract textual information from it. If the extracted text is insufficient, we leverage GPT to predict the semantic information of the clicked element.

4. Trajectory Visualization: We design a user-friendly interface to ensure a seamless annotation experience. For each action, we display its description, a corresponding video clip, and the a11y tree. Additionally, we provide the full video of the entire trajectory for better context.

5. Verification and Administration Systems: To ensure the quality of the collected data, we develop verification and administration systems that streamline the process of validating annotations and maintaining dataset integrity.

C.3.2 Privacy Protection

We implemented a multi-layer privacy protection framework in our data collection process. First, annotators must agree to a consent form that clearly states the scope of data collection, including screen recordings, actions, and system information. The form explicitly prohibits recording private or sensitive information. The tool is designed with privacy-first principles: no data is transmitted to servers without manual upload by annotators, and annotators can review all collected data (including videos, actions, and accessibility tree structures) before submission. We further ensure privacy through a two-stage verification process: manual review by internal team members during task verification, and automated examination of the task trajectory using GPT-4o during post-processing. Tasks containing private information are rejected immediately. Detailed privacy protection protocols are provided in the Appendix C.3.2.

1. GPT-Based Privacy Analysis

Data Ingestion: The system loads task descriptions and step-by-step user actions (*Observations*, *Thoughts*, *Action Descriptions*, etc.) from JSON. These records provide details of users’ intent, the interface elements users interacted with, and any textual or visual cues relevant to the task.

GPT Inference: The script calls OpenAI’s API with a carefully structured prompt, requesting GPT to produce a privacy classification in one of four levels: *None*, *Low*, *Medium*, or *High*. By passing the user’s detailed action steps and observations to GPT, the system gathers a structured output that includes an explicit `privacy_sensitivity` label.

GPT accelerates the classification process, effectively scanning large volumes of user trajectories with consistent logic. It can identify and label potentially sensitive fields or behaviors with minimal human oversight. However, GPT’s inherent reliance on keywords sometimes causes it to mark a scenario as *High*, even if the user simply viewed (but did not fill) a password field. Such errors underscore the necessity of post-processing or additional contextual checks to distinguish between *potential* versus *actual* private data entry.

2. Human Verification Human reviewers examine the same recordings or textual records to determine whether genuinely confidential information was entered. For example: If only an email address is mentioned, human reviewers might label it as *Low* or *None*. If a direct password or banking details appear, human reviewers assign *High*.

In “false alarm” cases (for example, a password field is displayed but left blank), humans typically label them as *None*. This nuanced reasoning often yields higher precision but comes at the cost of increased labor and time. Furthermore, variability in human judgments may occur if guidelines are not strictly enforced.

D OSWorld-Verified Results Detail

Table 10: Evaluation results of OPENCUA MODELS on **OSWorld-Verified**. OPENCUA-32B and OPENCUA-7B have been run for 3 turns. Pass@3 success rate is also calculated.

Model	Setting	#Step	Succ. Rate	Succ. Len	Fail Len	Calc	Impr	Writer	Chrome	VLC	TB	OS	GIMP	VSCode	Multi	Total Succ.
OPENCUA-32B	Turn 1	15	28.29%	7.34	12.79	7	14	8	17	4	6	10	15	10	10	101
	Turn 2	15	30.56%	7.34	12.85	5	15	8	21	5	8	13	14	12	9	110
	Turn 3	15	30.28%	7.31	12.67	7	14	11	18	4	6	13	14	14	8	109
	Avg.	15	29.71%	7.33	12.77	6.33	14.33	9.00	18.67	4.33	6.67	12.00	14.33	12.00	9.00	106.67
	Pass@3	15	37.34%	7.83	12.85	9	19	11	25	6	8	13	18	14	12	135
OPENCUA-32B	Turn 1	50	33.89%	10.40	23.79	6	18	9	20	4	9	11	19	10	16	122
	Turn 2	50	33.43%	10.53	24.62	7	16	10	19	5	8	12	18	13	12	120
	Turn 3	50	35.28%	11.29	24.42	6	20	9	21	4	7	14	17	15	14	127
	Avg.	50	34.20%	10.74	24.28	6.33	18.00	9.33	20.00	4.33	8.00	12.33	18.00	12.67	14.00	123.00
	Pass@3	50	45.58%	122.33	24.34	11	22	12	29	6	9	15	23	17	21	165
OPENCUA-32B	Turn 1	100	33.89%	12.86	32.52	11	15	8	16	6	7	12	18	12	17	122
	Turn 2	100	35.00%	10.86	29.90	8	18	8	19	5	6	11	18	16	17	126
	Turn 3	100	35.75%	11.13	29.47	7	20	9	20	6	8	15	16	15	12	128
	Avg.	100	34.88%	11.62	30.63	8.67	17.67	8.33	18.33	5.67	7.00	12.67	17.33	14.33	15.33	125.33
	Pass@3	100	45.10%	13.02	33.27	12	23	10	27	7	8	15	20	19	22	163
OPENCUA-7B	Turn 1	15	26.18%	7.70	12.73	5	17	6	17	5	8	7	13	10	6	94
	Turn 2	15	23.89%	6.97	12.77	4	14	7	13	3	6	10	12	10	7	86
	Turn 3	15	23.06%	6.94	12.92	4	12	5	15	4	6	8	10	10	9	83
	Avg.	15	24.38%	7.20	12.81	4.33	14.33	6.00	15.00	4.00	6.67	8.33	11.67	10.00	7.33	87.67
	Pass@3	15	30.74%	7.77	12.82	6	19	7	20	5	8	11	13	11	11	111
OPENCUA-7B	Turn 1	50	28.81%	10.06	24.14	7	13	8	18	4	7	11	13	10	13	104
	Turn 2	50	27.73%	8.93	24.77	5	17	8	18	5	6	10	10	11	9	99
	Turn 3	50	27.93%	8.12	24.34	7	16	7	17	5	6	9	11	11	11	100
	Avg.	50	28.16%	9.04	24.42	6.33	15.33	7.67	17.67	4.67	6.33	10.00	11.33	10.67	11.00	101.00
	Pass@3	50	35.75%	9.91	24.69	9	20	8	22	5	7	13	15	12	18	129
OPENCUA-7B	Turn 1	100	27.30%	9.79	28.71	4	16	6	18	5	6	10	14	10	9	98
	Turn 2	100	26.04%	9.68	27.55	5	13	6	18	5	6	10	11	10	10	94
	Turn 3	100	26.67%	9.61	27.20	6	16	7	14	5	6	10	12	11	9	96
	Avg.	100	26.67%	9.69	27.82	5.00	15.00	6.33	16.67	5.00	6.00	10.00	12.33	10.33	9.33	96.00
	Pass@3	100	36.48%	11.73	30.07	7	22	10	23	5	7	13	17	12	16	132
OPENCUA-A3B	Turn 1	15	16.90%	7.66	12.36	1	9	5	12	4	3	4	10	8	5	61
	Turn 1	50	19.94%	9.10	21.13	1	11	8	10	2	1	9	14	10	6	72
	Turn 1	100	17.73%	10.77	26.60	3	12	4	12	2	3	3	14	9	2	64
OPENCUA-QWEN2-7B	Turn 1	15	19.94%	7.25	12.74	4	11	8	9	5	6	4	10	12	3	72
	Turn 1	50	20.61%	8.54	21.28	3	12	10	15	5	7	2	6	10	4	74
	Turn 1	100	23.06%	9.73	26.19	5	14	10	12	4	5	6	10	11	6	83

E Error Study

After inspecting our online evaluation results, we group the failures into the following categories:

1. Insufficient task knowledge Foundation models may lack domain-specific GUI knowledge or the procedural know-how to finish a task. Each application has its own UI conventions and operation logic, and some tasks require specialized skills (e.g., spreadsheet formulas).

- Example 1: “I have a lookup table for the officers of each branch. Please fill the second table using VLOOKUP.” The agent does not know the VLOOKUP function and therefore fails.
- Example 2: “Fill all the blank cells with the value in the cell above.” The agent does not know the bulk-fill feature and instead edits cells one by one.

2. High-precision grounding errors Tasks that demand pixel-accurate actions frequently fail.

- Example: “Change the 2 in ‘H2O’ to a subscript.” The agent must precisely drag-select only the “2,” but often selects extra characters. Such fine-grained, letter-level grounding data are also hard to synthesize.

3. Action repetition When an incorrect action has no observable effect, the agent may keep predicting the same incorrect step. Occasionally it recovers, but often it loops indefinitely.

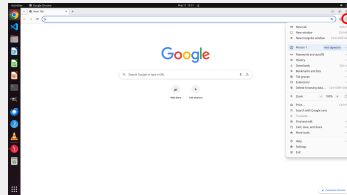
4. Termination misjudgment Sometimes the agent fails to notice that the task is already complete and continues acting, causing failure due to excessive extra actions. In other cases, it incorrectly assumes success and terminates prematurely.

5. Long-horizon task failures OSWorld includes tasks requiring >30–50 gold actions. Maintaining coherent context over so many steps remains challenging. For example: “Organize my desktop by placing academic papers in ‘Paper_reading’, coding projects in ‘Projects’, and everything else in ‘Miscellaneous’ . For files without clear names, determine the category by content.”

6. Insufficient error perception and recovery Although the agent can detect some mistakes and reflect, the agent is still not good at perceive error as human does. For example, high-precision edits still pose problems. It may insert text one character off yet judge the action correct, and it often lacks a reliable strategy to undo and retry after mistakes.

F OSWorld Case Example

The trajectory outlines the process of installing a manually developed Chrome extension located in the Desktop directory into the Google Chrome browser for testing or usage purposes. Notably, **the step labeled in red** highlights the agent's capacity for reflection and error correction—it initially diverges into an incorrect path, subsequently recognizes the mistake, and successfully reorients itself to proceed with the correct course of action.



Step 1

Thought: I need to install a Chrome extension that's located in the Desktop directory. Looking at the current state, I'm on the Google homepage in Chrome. To install an extension, I need to access the Chrome Web Store or use the extensions menu.

I can see that there's a "Web Store" shortcut on the Google homepage which could be useful. However, I also notice the Chrome menu button (three vertical dots) in the top-right corner of the browser. This menu would give me access to more Chrome settings and features, including extensions.

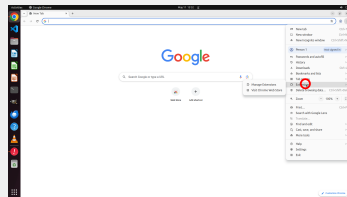
Since I need to install a local extension file, the most direct approach would be to: 1. Click on the Chrome menu (three dots) 2. Navigate to "More tools" > "Extensions" or directly to "Chrome Web Store" 3. Upload the extension file from the Desktop.

The Chrome menu button in the top-right corner is the logical next action because it provides access to all Chrome settings and features, including extensions management. Clicking this button will open a dropdown menu with various options, including those related to extensions.

I anticipate that after clicking the menu button, I'll see options that will allow me to navigate to the extensions page or Chrome Web Store, where I can then upload the extension file from the Desktop.

Action: Click on the three-dot menu button in the top-right corner of the Chrome browser window to open the Chrome menu.

Code: pyautogui.click(x=0.988, y=0.081)



Step 2

Thought: After clicking on the three-dot menu in the previous step, the Chrome menu has successfully opened, displaying various options. This is exactly what I expected to happen, so we're making good progress toward our goal of installing a Chrome extension from the Desktop. Looking at the current state, I can see that the Chrome menu is now open with all its options. To install a Chrome extension, I need to access the Extensions page or use the Web Store. Based on the goal of installing an extension from the Desktop directory, I should look for the Extensions option in this menu.

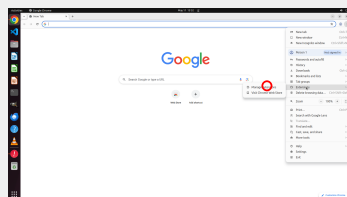
I can see that there is an "Extensions" option in the menu, which is exactly what I need. Clicking on this option will take me to Chrome's Extensions page, where I can then load an unpacked extension from the Desktop directory.

The logical next action would be to click on the "Extensions" option in the Chrome menu. This will navigate to the Extensions management page where I can then find an option to load an unpacked extension. From there, I'll be able to browse to the Desktop directory and select the extension file for installation.

By clicking on the Extensions option, I expect to be taken to the Extensions management page where I'll see a list of currently installed extensions and options to manage them, including an option to "Load unpacked" which I'll need to use to install the extension from the Desktop.

Action: Click on the "Extensions" option in the Chrome menu to access the extensions management page.

Code: pyautogui.click(x=0.867, y=0.406)



Step 3

Thought: After clicking on the "Extensions" option in the Chrome menu, I can see that the menu has expanded to show more options related to extensions. This is the expected result of the previous action.

Looking at the current state, I can see that there are two main options related to extensions: 1. "Manage Extensions" - This would take me to the page where I can manage currently installed extensions 2. "Visit Chrome Web Store" - This would take me to the Chrome Web Store where I can download extensions from the official store

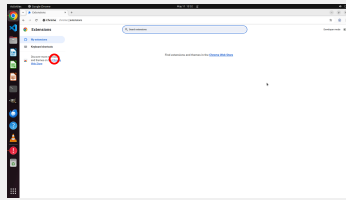
Since our goal is to install a Chrome extension that is located on the Desktop directory, I need to access the extensions management page first. From there, I would be able to access the "Load unpacked" button which allows installing extensions from local files rather than from the Chrome Web Store.

The logical next action would be to click on "Manage Extensions" since this is the option that will take me to the extensions management page where I can install the extension from the Desktop. This is the most direct path to achieving our goal of installing a locally developed extension.

By clicking on "Manage Extensions," I expect to be taken to a page that shows all currently installed extensions and provides options to add new extensions, including the "Load unpacked" option that I'll need to use to install the extension from the Desktop directory.

Action: Click on the "Manage Extensions" option in the Chrome menu to access the extensions management page.

Code: pyautogui.click(x=0.755, y=0.403)



Step 4

Thought: After clicking on "Manage Extensions" in the previous step, I can see that I've successfully navigated to the Chrome extensions management page. This is exactly where I need to be to continue with the installation of the new Chrome extension.

Looking at the current state, I can see that I'm on the `chrome://extensions` page which shows the list of currently installed extensions. However, I don't see any extensions listed yet, which suggests either there are no extensions installed or the page needs to be refreshed to show them.

To install a new extension manually from a local file on the Desktop, I need to: 1. Access the Chrome Web Store or a similar interface where I can upload or load an extension from a file 2. Navigate to the extension file location (Desktop) 3. Install the extension

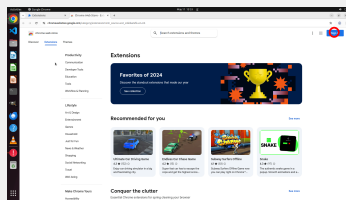
I notice there are a few options available to me: - I could click on "My extensions" to see if there are any installed extensions - I could click on "Keyboard shortcuts" to configure shortcuts - I could click on the "Chrome Web Store" link to browse the store - I could click on the "Chrome" link in "Discover more extensions and themes on the Chrome Web Store"

Since my goal is to install a locally developed extension, I need to find an option that would allow me to load an extension from a file. The "Chrome Web Store" link seems to be the most logical next step, as it would likely provide access to extension management functions including loading local extensions.

Clicking on the "Chrome" link in the text "Discover more extensions and themes on the Chrome Web Store" would take me to the Chrome Web Store, where I might find options to access developer tools or load extensions from local files. This seems like the most logical next action to progress toward my goal of installing the extension from the Desktop directory.

Action: Click on the "Chrome" link in the text "Discover more extensions and themes on the Chrome Web Store"

Code: `pyautogui.click(x=0.138, y=0.284)`



Step 5

Thought: After clicking on the "Chrome" link in the Extensions management page, I've been taken to the Chrome Web Store. This is a good start, but I notice that I'm currently in the Extensions category of the Chrome Web Store, which is for browsing and downloading extensions from the store, not for installing local extensions.

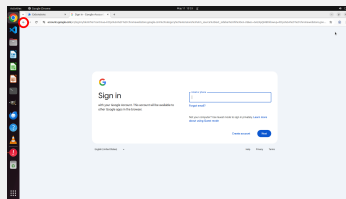
To install a Chrome extension from a local file (located in the Desktop directory as mentioned in the goal), I need to access the Chrome Web Store's developer page or use a different approach. Looking at the current state, I need to navigate to a different section that allows for installing unpacked extensions.

I can see that there's a "Sign In" button in the top-right corner of the page. Signing in to Chrome is often necessary for extension development and installation. Additionally, there might be a way to access developer options or a special section for installing unpacked extensions after signing in. Given the goal of installing a locally developed Chrome extension from the Desktop directory, the next logical step would be to sign in to Chrome. This would likely give me access to more advanced options for extension installation, including the ability to load an unpacked extension from a local directory.

After signing in, I expect to either find a direct option to load an unpacked extension or need to upload the extension file from the Desktop. Signing in is a necessary step in the process of manually installing a Chrome extension from a local directory.

Action: Click on the "Sign In" button in the top-right corner of the Chrome Web Store page.

Code: `pyautogui.click(x=0.954, y=0.131)`



Step 6

Thought: After clicking the "Sign In" button in the Chrome Web Store, I've been redirected to a Google sign-in page. This makes sense as many Chrome extensions require a signed-in profile to be installed properly. The sign-in page is asking for Google account credentials to proceed.

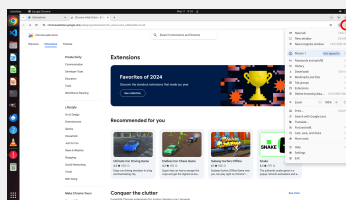
Looking at my progress toward the goal of installing a Chrome extension manually from the Desktop directory, I need to: 1. Access the Chrome Web Store (completed) 2. Sign in to Chrome (in progress) 3. Navigate to the extension file on the Desktop 4. Install the extension manually I'm currently at step 2, where I need to sign in to a Google account to proceed. However, I notice that I might not need to complete the entire sign-in process. Since the goal is to install a locally downloaded extension, I might be able to go back and find an option to load an extension from a local file without fully signing in.

Looking at the browser interface, I see the back button in the top-left corner of the browser. Clicking this back button would allow me to return to the Chrome Web Store page where I might find alternative options for installing an extension from a local file, such as a "Load unpacked" button that's typically available when you're in the extensions management page.

The logical next action would be to click the back button to return to the Chrome Web Store. This would allow me to potentially find a more direct way to install the extension from the Desktop without completing the sign-in process. After clicking back, I expect to return to the Chrome Web Store page where I can look for options related to loading local extensions.

Action: Click the back button in the top-left corner of the Chrome browser window to return to the previous page.

Code: `pyautogui.click(x=0.048, y=0.086)`



Step 7

Thought: After clicking the back button, I've successfully returned to the Chrome Web Store page from the Google sign-in page. This was the right move since signing in wasn't necessary to install a local extension.

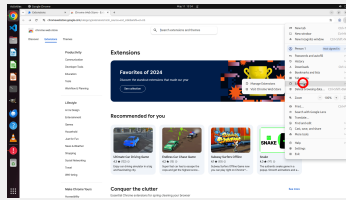
Now I need to find a way to install the extension from the Desktop directory. Looking at the current state of the Chrome Web Store page, I need to find an option that would allow me to load an extension from a local file. This functionality is typically available through an "More" menu or similar option on the Chrome Web Store page.

I can see that there's a three-dot menu button in the top-right corner of the Chrome browser window. This is likely where I'll find additional options for managing extensions, including the ability to load an extension from a local file. Clicking this button should reveal a dropdown menu with various options, potentially including "More Extensions" or "Load unpacked extension" which would allow me to navigate to the Desktop directory and select the extension file.

The logical next action is to click on this three-dot menu button in the top-right corner of the browser. This should open a menu with additional options related to extensions, which will help me progress toward manually installing the extension from the Desktop directory.

Action: Click on the three-dot menu button in the top-right corner of the Chrome browser window to access additional browser options.

Code: `pyautogui.click(x=0.988, y=0.083)`



Step 8

Thought: After clicking the three-dot menu in the top-right corner of the Chrome browser, a dropdown menu has appeared with various browser options. This is exactly what I was expecting to see, as this menu provides access to Chrome's core functionality including extensions management. Looking at my progress toward the goal of installing a Chrome extension manually from the Desktop directory, I've successfully: 1. Opened the Chrome menu 2. Accessed the Extensions option 3. Navigated to the Chrome Web Store 4. Attempted to sign in (but determined it wasn't necessary) 5. Opened the Chrome menu again to access more options

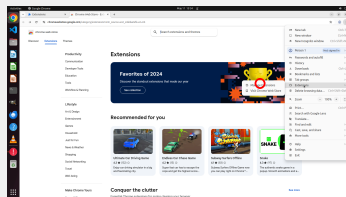
Now I need to find the best way to install the extension from the Desktop. Since I already know the extension file is on the Desktop, I should look for an option that would allow me to load an unpacked extension or access the Desktop files.

Looking at the dropdown menu, I can see an "Extensions" option with a submenu indicator (>). This is exactly what I need to click next because: 1. It will take me to the Extensions management page or provide extension-related options 2. From there, I should be able to find an option to load an unpacked extension 3. This is the standard way in Chrome to manage extensions, including loading them from local files

Clicking on the "Extensions" menu item will likely show me additional options such as "Load unpacked" which would allow me to navigate to the Desktop and select the extension folder. This is the most direct path to manually installing the extension from the Desktop as required.

Action: Click on the "Extensions" option in the Chrome menu to access extension management functions.

Code: `pyautogui.click(x=0.863, y=0.406)`



Step 9

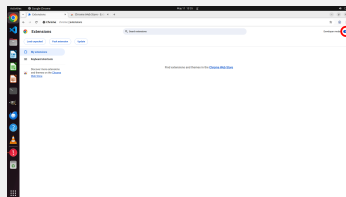
Thought: After clicking on the "Extensions" option in the Chrome menu, a submenu has appeared with two options: "Manage Extensions" and "Visit Chrome Web Store". This is exactly what I need to continue with the installation process.

Looking at my progress toward installing the Chrome extension: 1. I opened the Chrome menu and accessed Extensions 2. I went to the Chrome Web Store's Extensions page 3. I attempted to sign in but decided to go back 4. I opened the Chrome menu again and clicked on Extensions 5. Now I'm seeing a submenu with extension-related options

To install a local extension from the Desktop directory, I need to access the extension management interface. The "Manage Extensions" option is exactly what I need, as it will allow me to access the extension installation functionality. This is where I'll be able to load an unpacked extension from a local directory. The "Manage Extensions" option is the first item in the submenu that has appeared. Clicking on this option will take me to the extensions management page where I can find the option to add an unpacked extension. From there, I'll be able to navigate to the Desktop directory and select the extension file to complete the installation. The logical next action is to click on "Manage Extensions" in the submenu. This will take me to the extensions management page where I can proceed with installing the extension from the Desktop directory.

Action: Click on the "Manage Extensions" option in the Chrome menu submenu.

Code: `pyautogui.click(x=0.738, y=0.401)`



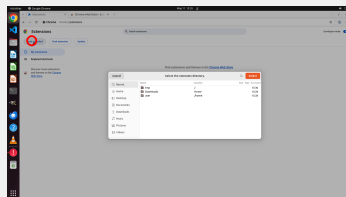
Step 10

Thought: After clicking on "Manage Extensions" in the Chrome menu, I've been taken to the Chrome extensions management page. This is exactly where I need to be to continue with the installation of the new Chrome extension.

Looking at my progress toward the goal of installing a Chrome extension manually from the Desktop directory: 1. I've successfully accessed the Chrome browser 2. I've navigated to the extensions management page 3. I now need to find a way to load the extension from the Desktop. The extensions page shows several options, including "My extensions" and "Keyboard shortcuts," but I don't see a direct option to upload or add an extension from a file. However, I notice the "Developer mode" toggle in the top-right corner of the page. This is exactly what I need, as Developer mode typically provides additional options for developers, including the ability to load unpacked extensions. To install a manual extension, I need to enable Developer mode first. Once Developer mode is enabled, I should get additional options that will allow me to load an unpacked extension from a local directory like the Desktop. The logical next action is to click on the "Developer mode" toggle switch in the top-right corner of the page. This will enable developer options and should reveal additional controls that will allow me to load the extension from the Desktop directory. After clicking this toggle, I expect to see new buttons appear that will let me access the file system and select the extension file.

Action: Click on the "Developer mode" toggle switch in the top-right corner of the page to enable developer options.

Code: `pyautogui.click(x=0.988, y=0.131)`



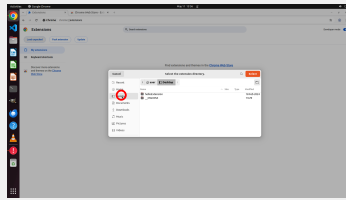
Step 11

Thought: After clicking on the "Developer mode" toggle switch, it has successfully been enabled as shown by the blue toggle in the top-right corner of the page. This has revealed additional functionality that wasn't visible before, including the "Load unpacked" button which is exactly what we need to install our extension manually. Now that Developer mode is enabled, I can see the "Load unpacked" button which allows us to install extensions from local files rather than from the Chrome Web Store. This is perfect for our goal since we need to install an extension that's located in the Desktop directory. Looking at the available options, I can see: 1. "Load unpacked" button - This is what we need to install our local extension 2. "Pack extension" button - Used for creating a packaged version of an extension 3. "Update" button - For updating existing extensions 4. The sidebar navigation showing "My extensions" and "Keyboard shortcuts" To install the extension from the Desktop directory, the logical next action is to click on the "Load unpacked" button. This will open a file browser dialog that will allow us to navigate to the Desktop and select the extension folder. Once we select the folder, Chrome will install the extension from the unpacked files. Clicking the "Load unpacked" button is the most appropriate action because: 1. It directly addresses our goal of installing a locally developed extension 2. It's the standard way to install extensions from local files in Chrome 3. The Developer mode is already enabled, which is a prerequisite for using this feature

After clicking this button, I expect a file browser dialog to appear, allowing us to navigate to the Desktop and select the extension folder.

Action: Click on the "Load unpacked" button to open a file browser dialog for selecting the extension folder from the Desktop.

Code: `pyautogui.click(x=0.072, y=0.179)`



Step 12

Thought: After clicking the "Load unpacked" button, a file browser dialog has opened as expected. This dialog allows me to navigate to the location where the extension is stored, which according to our goal should be in the Desktop directory.

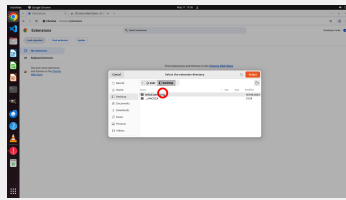
Looking at the current state, I can see that the file browser dialog is showing the contents of what appears to be the home directory, with folders like tmp, Downloads, and user visible. However, our goal is to install a Chrome extension that's located in the Desktop directory. Therefore, I need to navigate to the Desktop folder.

I notice that the Desktop folder is conveniently listed in the left navigation panel of the file browser. Since we need to access the Desktop to find the extension files, clicking on the Desktop folder in the left navigation panel is the logical next step. This will change the view to show the contents of the Desktop directory, where we should be able to find the extension files.

Once I click on Desktop, I expect the main panel to update and show all files and folders located on the Desktop. From there, I'll need to locate the specific extension folder and select it to complete the installation process.

The logical next action is to click on the "Desktop" option in the left navigation panel of the file browser dialog to navigate to the Desktop directory where our extension is located. **Action:** Click on the "Desktop" option in the left navigation panel of the file browser dialog.

Code: pyautogui.click(x=0.333, y=0.470)



Step 13

Thought: After clicking on the "Desktop" option in the left navigation panel, the file browser dialog has now displayed the contents of the Desktop directory. This is exactly what we needed to proceed with installing the Chrome extension.

Looking at the current state, I can see that the Desktop directory contains two items: a folder named "helloExtension" and another folder named "_MACOSX". Since we're looking to install a Chrome extension, the "helloExtension" folder is most likely the extension we need to install. Chrome extensions typically follow a specific folder structure with an "manifest.json" file at the root, and "helloExtension" appears to be a properly named extension folder. Now that I've located what appears to be the extension folder, the next logical step would be to select this folder to proceed with the installation. According to the goal, we need to install a Chrome extension located in the Desktop directory, and we've now found what seems to be that extension.

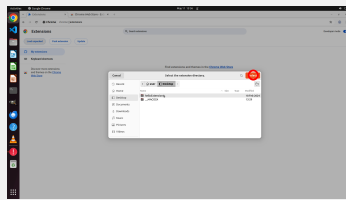
I have several options at this point: 1. Click on the "helloExtension" folder to select it 2. Click the "Select" button (which would likely select the currently highlighted item) 3. Navigate into the folder to confirm its contents

Since the "helloExtension" folder is the most likely candidate for the extension we need to install, and it's already visible in the file browser, clicking on this folder to select it is the most logical next action. This will highlight the folder and prepare it for selection, after which I can click the "Select" button to complete the installation process.

I expect that after clicking on the "helloExtension" folder, it will become highlighted, indicating it's selected. Then I can proceed to click the "Select" button to load this unpacked extension into Chrome.

Action: Click on the "helloExtension" folder in the file browser dialog to select it for installation.

Code: pyautogui.click(x=0.454, y=0.456)



Step 14

Thought: After clicking on the "helloExtension" folder in the file browser dialog, I can see that the folder is now selected. This is exactly what we needed to do to identify the extension we want to install. The file browser dialog shows we're in the Desktop directory, and we can see the "helloExtension" folder that contains our newly developed Chrome extension.

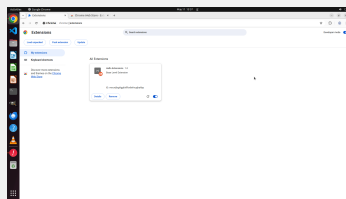
Now that we've selected the "helloExtension" folder, the next logical step is to confirm this selection by clicking the "Select" button in the file browser dialog. This button is prominently displayed in orange in the top-right corner of the dialog. Clicking this button will finalize our selection and proceed with the installation process.

The installation workflow so far has been: 1. We enabled Developer mode in Chrome 2. We clicked "Load unpacked" to open the file browser 3. We navigated to the Desktop directory 4. We selected the "helloExtension" folder

The final step to complete the installation is to click the "Select" button, which will tell Chrome to load the unpacked extension from the selected folder. After clicking this button, Chrome should install the extension and possibly show a confirmation message or automatically add the extension to the browser. I expect that after clicking the "Select" button, the file browser dialog will close, and we'll return to the Extensions page in Chrome where we might see the newly installed extension appear in the list of installed extensions, or possibly see a notification about the successful installation.

Action: Click the "Select" button in the top-right corner of the file browser dialog to confirm the selection of the helloExtension folder and complete the installation process.

Code: pyautogui.click(x=0.718, y=0.358)



Step 15

Thought: After clicking the "Select" button in the file browser dialog, the extension installation process has completed successfully. The dialog has closed and we've been returned to the Extensions management page where we can now see that the "Hello Extensions" extension has been installed. The extension appears with its icon, version number (1.0), and description ("Base Level Extension"). Looking at our progress toward the goal of installing the Chrome extension: 1. We enabled Developer mode in the Extensions page ✓ 2. We clicked the "Load unpacked" button which opened the file browser dialog ✓ 3. We navigated to the Desktop directory ✓ 4. We selected the "helloExtension" folder ✓ 5. We clicked the "Select" button to confirm the installation ✓ 6. The extension has now been successfully installed ✓ The task has been completed successfully!

The extension is now installed and appears in the Extensions management page. I can see that it's enabled (the toggle switch is turned on). If we wanted to configure any specific settings for this extension, we could click on the "Details" button which would likely show more options for the extension. Since our goal was to install the extension manually from the Desktop directory, and that has been accomplished, there's no further action needed for the basic installation task. The extension is now available for use in Chrome.

Action: Terminate the process with success status as the extension installation is complete.

Code: computer.terminate(status='success')

G AgentNet Training Data Example

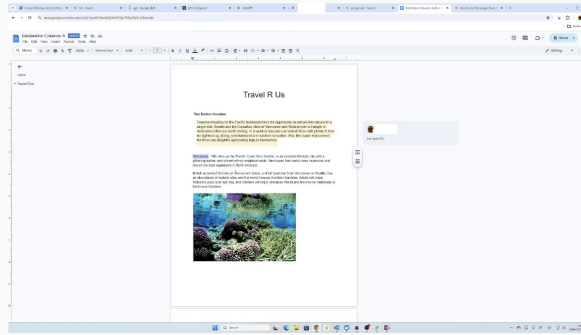
G.1 L1 Data Example

L1 data includes CoT-L1(Action) in its inner monologue. The assistant's final output will be included in the loss calculation (green color content).

L1 Data Example

```
[
  {
    "role": "system",
    "content": "You are a GUI agent. You are given a task and a screenshot of the screen. You need to perform a series of pyautogui actions to complete the task. For each step, provide your response in this format: Action: Provide clear, concise, and actionable instructions: - If the action involves interacting with a specific target: - Describe target explicitly without using coordinates - Specify element names when possible (use original language if non-English) - Describe features (shape, color, position) if name unavailable - For window control buttons, identify correctly (minimize, maximize, close) - if the action involves keyboard actions like 'press', 'write', 'hotkey': - Consolidate repetitive keypresses with count - Specify expected text outcome for typing actions Finally, output the action as PyAutoGUI code or the following functions:- {"name": "computer.triple_click", "description": "Triple click on the screen", "parameters": {"type": "object", "properties": {"x": {"type": "number", "description": "The x coordinate of the triple click"}, "y": {"type": "number", "description": "The y coordinate of the triple click"}}, "required": ["x", "y"]}- {"name": "computer.terminate", "description": "Terminate the current task and report its completion status", "parameters": {"type": "object", "properties": {"status": {"type": "string", "enum": ["success", "failure"]}, "description": "The status of the task"}}, "required": ["status"]}"
  },
  {
    "role": "assistant",
    "content": "## Step 1:
    ## Action: Click on the \"Destination Columns-A...\" document thumbnail to open the document.

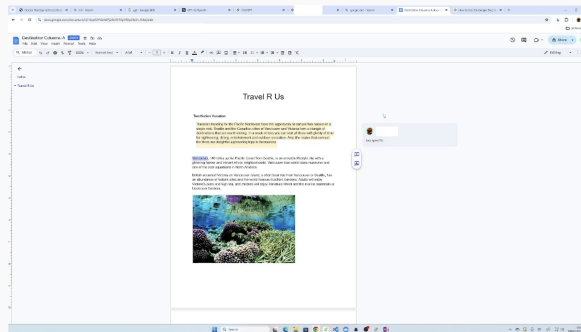
    # Step 2:
    ## Action: Select the word \"Travelers\" at the beginning of the second paragraph by clicking at the start of the word and dragging to the end of it."
  },
  {
    "role": "user",
    "image":
```



```

},
{
  "role": "assistant",
  "content": "# Step 3:
  ## Action:Press Ctrl+C to copy the selected word \"Travelers\"
},
{
  "role": "user",
  "image":

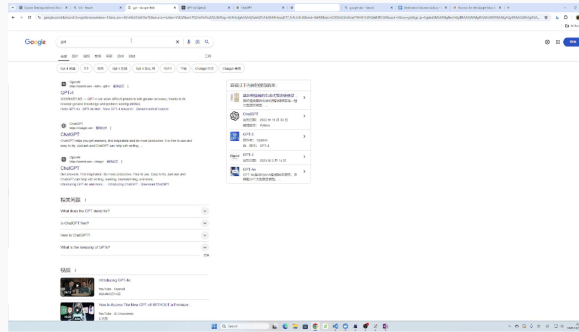
```



```

},
{
  "role": "assistant",
  "content": "# Step 4:
  ## Action:Click on the Google search tab near the top of the browser
  window."
},
{
  "role": "user",
  "image":

```



```

},
{
  "role": "user",
  "content": "# Task Instruction:
  Could you open the document 'Destination Columns-A', find the first
  word of the second paragraph, look up what that word means on
  Wikipedia, and then add that Wikipedia description as a footnote to
  that word in the document?
  Please generate the next move according to the screenshot, task
  instruction and previous steps (if provided).",
},
{
  "role": "assistant",
  "content": "# Step 5:
  ## Action:Click on the Google search box where "gpt" is currently
  displayed to prepare to enter a new search query.
  ## Code:''pythonpyautogui.click(x=0.157, y=0.1229)''",
}

```

G.2 L2 Data Example

L2 data includes CoT-L2(Thought, Action) in its inner monologue. The assistant's final output will be included in the loss calculation.

L2 Data Example

```

[
  {
    "role": "system",

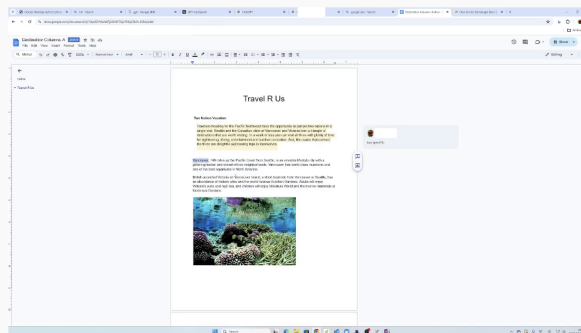
```

```

"content": "You are a GUI agent. You are given a task and a
screenshot of the screen. You need to perform a series of pyautogui
actions to complete the task. For each step, provide your response in
this format: Thought: - Step by Step Progress Assessment: -
Analyze completed task parts and their contribution to the overall
goal - Reflect on potential errors, unexpected results, or
obstacles - If previous action was incorrect, predict a logical
recovery step - Next Action Analysis: - List possible next
actions based on current state - Evaluate options considering
current state and previous actions - Propose most logical next
action - Anticipate consequences of the proposed action - For
Text Input Actions: - Note current cursor position -
Consolidate repetitive actions (specify count for multiple
keypresses) - Describe expected final text outcome - Use
first-person perspective in reasoningAction: Provide clear, concise,
and actionable instructions: - If the action involves interacting
with a specific target: - Describe target explicitly without using
coordinates - Specify element names when possible (use original
language if non-English) - Describe features (shape, color,
position) if name unavailable - For window control buttons,
identify correctly (minimize, maximize, close) - if the action
involves keyboard actions like 'press', 'write', 'hotkey': -
Consolidate repetitive keypresses with count - Specify expected
text outcome for typing actions Finally, output the action as
PyAutoGUI code or the following functions:- {"name":
"computer.triple_click", "description": "Triple click on the screen",
"parameters": {"type": "object", "properties": {"x": {"type":
"number", "description": "The x coordinate of the triple click"}, "y":
{"type": "number", "description": "The y coordinate of the triple
click"}}, "required": ["x", "y"]}- {"name": "computer.terminate",
"description": "Terminate the current task and report its completion
status", "parameters": {"type": "object", "properties": {"status":
{"type": "string", "enum": ["success", "failure"], "description":
"The status of the task"}}, "required": ["status"]}}"
},
{
"role": "assistant",
"content": "# Step 1:
## Action: Click on the "Destination Columns-A..." document thumbnail
to open the document.

# Step 2:
## Action: Select the word "Travelers" at the beginning of the second
paragraph by clicking at the start of the word and dragging to the
end of it."
},
{
"role": "user",
"image":

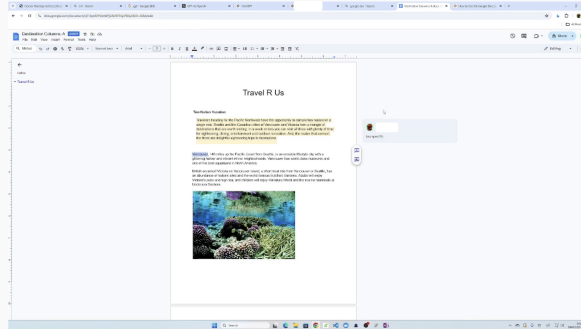
```



```

},
{
  "role": "assistant",
  "content": "# Step 3:  
## Action:Press Ctrl+C to copy the selected word \"Travelers\""  
},
{
  "role": "user",
  "image":

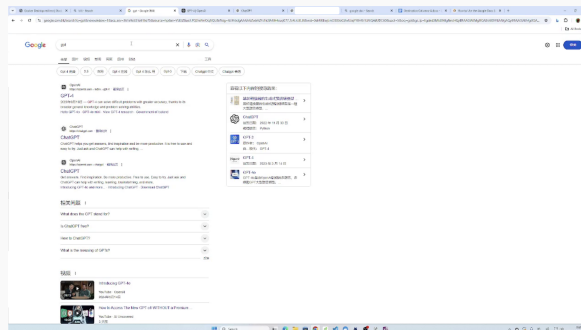
```



```

},
{
  "role": "assistant",
  "content": "# Step 4:  
## Action:Click on the Google search tab near the top of the browser  
window."  
},
{
  "role": "user",
  "image":

```



```

},
{
  "role": "user",
  "content": "# Task Instruction:  
Could you open the document 'Destination Columns-A', find the first  
word of the second paragraph, look up what that word means on  
Wikipedia, and then add that Wikipedia description as a footnote to  
that word in the document?  
Please generate the next move according to the screenshot, task  
instruction and previous steps (if provided)."
```

```

    },
    {
      "role": "assistant",
      "content": "# Step 5:
## Thought:After clicking on the Google search tab, I can now see
that the browser has successfully switched to a Google search page.
This is a step forward in my task, as I need to search for the word
"Travelers" on Wikipedia to complete the footnote insertion.Currently,
I see that the search box contains "gpt" rather than the word
"Travelers" that I copied from the document. To proceed with my task,
I need to:1. Clear the current search term "gpt" from the search box2.
Enter "Travelers Wikipedia" or navigate directly to Wikipedia and
search there3. Find relevant information about "Travelers" on
Wikipedia4. Return to the Google Docs document and insert this
information as a footnoteMy progress on the overall task so far:1.
I've opened the "Destination Columns-A" document v2. I've identified
and selected the first word of the second paragraph ("Travelers") v3.
I've copied this word to the clipboard v4. I've switched to a browser
tab for searching v5. Now I need to clear the current search and look
up "Travelers" on Wikipedia6. Then I need to insert the Wikipedia
description as a footnoteLooking at the current state, I need to
click directly on the search box where "gpt" is currently displayed.
By clicking on the search box, I'll be able to select all the text,
delete it, and replace it with "Travelers Wikipedia" to search for
information about Travelers on Wikipedia. This is the most efficient
next action as it allows me to directly modify the search query
without any additional steps.After clicking on the search box, I
expect the text cursor to be positioned within the box, allowing me
to edit the current search term. I can then clear "gpt" and type
"Travelers Wikipedia" to continue with my search. This will help me
find the Wikipedia page for "Travelers" so I can obtain the
description needed for the footnote.
## Action:Click on the Google search box where "gpt" is currently
displayed to prepare to enter a new search query.
## Code:''pythonpyautogui.click(x=0.157, y=0.1229)''"
    }
  ]

```

G.3 L3 Data Example

L3 data includes CoT-L3(Observation, Thought, Action) in its inner monologue. The assistant's final output will be included in the loss calculation.

L3 Data Example

```

[
  {
    "role": "system",

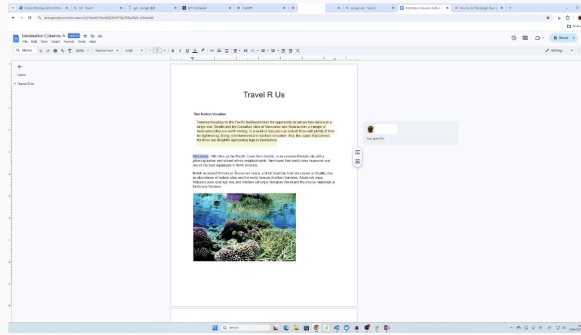
```

```

"content": "You are a GUI agent. You are given a task and a
screenshot of the screen. You need to perform a series of pyautogui
actions to complete the task. For each step, provide your response in
this format: Observation: - Describe the current computer state based
on the full screenshot in detail. - Application Context: - The
active application - The active window or page - Overall layout
and visible interface - Key Elements: - Menu items and toolbars
- Buttons and controls - Text fields and content - Dialog boxes
or popups - Error messages or notifications - Loading states
- Other key elements - Describe any content, elements, options,
information or clues that are possibly relevant to achieving the task
goal, including their name, content, or shape (if possible). Thought:
- Step by Step Progress Assessment: - Analyze completed task parts
and their contribution to the overall goal - Reflect on potential
errors, unexpected results, or obstacles - If previous action was
incorrect, predict a logical recovery step - Next Action Analysis:
- List possible next actions based on current state - Evaluate
options considering current state and previous actions - Propose
most logical next action - Anticipate consequences of the proposed
action - For Text Input Actions: - Note current cursor position
- Consolidate repetitive actions (specify count for multiple
keypresses) - Describe expected final text outcome - Use
first-person perspective in reasoning Action: Provide clear, concise,
and actionable instructions: - If the action involves interacting
with a specific target: - Describe target explicitly without using
coordinates - Specify element names when possible (use original
language if non-English) - Describe features (shape, color,
position) if name unavailable - For window control buttons,
identify correctly (minimize, maximize, close) - if the action
involves keyboard actions like 'press', 'write', 'hotkey': -
Consolidate repetitive keypresses with count - Specify expected
text outcome for typing actions Finally, output the action as
PyAutoGUI code or the following functions:- {"name":
"computer.triple_click", "description": "Triple click on the screen",
"parameters": {"type": "object", "properties": {"x": {"type":
"number", "description": "The x coordinate of the triple click"}, "y":
{"type": "number", "description": "The y coordinate of the triple
click"}}}, "required": ["x", "y"]}- {"name": "computer.terminate",
"description": "Terminate the current task and report its completion
status", "parameters": {"type": "object", "properties": {"status":
{"type": "string", "enum": ["success", "failure"], "description":
"The status of the task"}}, "required": ["status"]}}"
},
{
  "role": "assistant",
  "content": "## Step 1:
## Action: Click on the \"Destination Columns-A...\" document thumbnail
to open the document.

# Step 2:
## Action: Select the word \"Travelers\" at the beginning of the second
paragraph by clicking at the start of the word and dragging to the
end of it."
},
{
  "role": "user",
  "image":

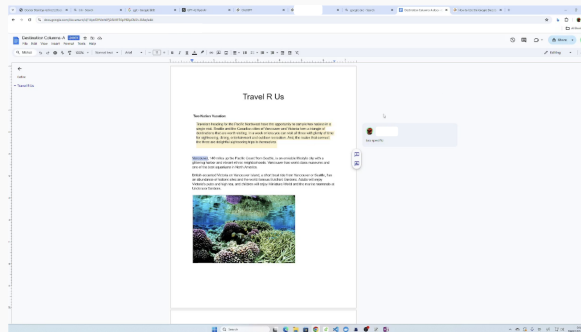
```



```

},
{
  "role": "assistant",
  "content": "# Step 3:
  ## Action:Press Ctrl+C to copy the selected word \"Travelers\"
},
{
  "role": "user",
  "image":

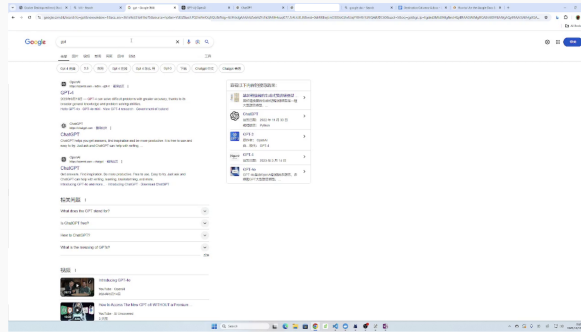
```



```

},
{
  "role": "assistant",
  "content": "# Step 4:
  ## Action:Click on the Google search tab near the top of the browser
  window."
},
{
  "role": "user",
  "image":

```

```

},
{
  "role": "user",
  "content": "# Task Instruction:
  Could you open the document 'Destination Columns-A', find the first
  word of the second paragraph, look up what that word means on
  Wikipedia, and then add that Wikipedia description as a footnote to
  that word in the document?
  Please generate the next move according to the screenshot, task
  instruction and previous steps (if provided).",
},
{
  "role": "assistant",

```

```

"content": "# Step 5:
## Observation:I'm looking at a Google search page where "gpt" has
been entered in the search box. The interface shows Google search
results for "gpt" with various tabs available at the top (All, Images,
Videos, Shopping, News, More). The search results show information
about GPT and related technologies. Multiple search results are
visible, including:- Information about GPT-4- ChatGPT (from
https://chatgpt.com)- Multiple websites with GPT-related content The
browser has multiple tabs open at the top, including a tab for
"Destination Columns-A" document, which appears to be a Google Docs
document that was previously being worked on. There's also a search
bar at the top with the query "gpt" entered.Below the search results
are some related questions in a FAQ section, such as "What does the
GPT stand for?" and "Is ChatGPT free?" among others.The Google search
interface includes various filters and options to refine the search,
including some buttons like "Gpt-4 " (GPT-4 relevance) and several
other filter options.
## Thought:After clicking on the Google search tab, I can now see
that the browser has successfully switched to a Google search page.
This is a step forward in my task, as I need to search for the word
"Travelers" on Wikipedia to complete the footnote insertion.Currently,
I see that the search box contains "gpt" rather than the word
"Travelers" that I copied from the document. To proceed with my task,
I need to:1. Clear the current search term "gpt" from the search box2.
Enter "Travelers Wikipedia" or navigate directly to Wikipedia and
search there3. Find relevant information about "Travelers" on
Wikipedia4. Return to the Google Docs document and insert this
information as a footnoteMy progress on the overall task so far:1.
I've opened the "Destination Columns-A" document v2. I've identified
and selected the first word of the second paragraph ("Travelers") v3.
I've copied this word to the clipboard v4. I've switched to a browser
tab for searching v5. Now I need to clear the current search and look
up "Travelers" on Wikipedia6. Then I need to insert the Wikipedia
description as a footnoteLooking at the current state, I need to
click directly on the search box where "gpt" is currently displayed.
By clicking on the search box, I'll be able to select all the text,
delete it, and replace it with "Travelers Wikipedia" to search for
information about Travelers on Wikipedia. This is the most efficient
next action as it allows me to directly modify the search query
without any additional steps.After clicking on the search box, I
expect the text cursor to be positioned within the box, allowing me
to edit the current search term. I can then clear "gpt" and type
"Travelers Wikipedia" to continue with my search. This will help me
find the Wikipedia page for "Travelers" so I can obtain the
description needed for the footnote.
## Action:Click on the Google search box where "gpt" is currently
displayed to prepare to enter a new search query.
## Code:''pythonpyautogui.click(x=0.157, y=0.1229)''''

```

```

}

```